

CYP2D6 DDGI Qualification

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Qualification Plan Release	https://github.com/Open-Systems-Pharmacology/DDGI-CYP2D6
OSP Version	12.2
Qualification Framework Version	3.5

This qualification report is filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-Qualification-Reports>

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1 Introduction

1.1 Objective

This qualification report evaluates the developed physiologically based pharmacokinetic (PBPK) drug-drug-gene interaction (DDGI) network for the ability to perform simulations with the intended purpose to predict cytochrome P450 2D6 (**CYP2D6**)-mediated drug-drug interactions (DDIs) and DDGIs.

To demonstrate the level of confidence, the predictive performance is assessed using a network of PBPK models for selected CYP2D6 perpetrators, victims, and genotype-dependent interaction scenarios together with clinical DD(G)I data from published studies. The models are whole-body PBPK models and allow dynamic interaction simulations in tissues expressing the relevant enzymes or transporters.

The respective *qualification plan* to produce this *qualification report* is transparently documented and provided open-source (<https://github.com/Open-Systems-Pharmacology/DDGI-CYP2D6>). The same applies for all presented PBPK models including *evaluation reports* on model building and evaluation of each model (<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library>).

Evaluation reports including descriptions on model building and detailed evaluations of the included models are documented separately (see [Section 1.2](#)).

Please refer to the [Appendix](#) to learn more details:

- An overview over the Open Systems Pharmacology (OSP) Suite is given in chapter [Section 5.1](#)
- [Section 5.2](#) shows the implementation of the underlying mathematical equations for DDIs in the OSP suite.
- A detailed general description of the performed qualification workflow (*qualification plan*, *qualification report*, etc.) can be found in chapter [Section 5.3](#).

2 DDI Evaluations

The following section shows the correlations between observed and model-predicted AUC_{last} and C_{max} ratios, respectively.

Specifically, the PBPK model performance for the PK parameters **AUC ratio (AUCR)** and **C_{max} ratio (CmaxR)** is assessed via:

- predicted (*Pred*) vs. observed (*Obs*) plots
- *Pred/Obs* vs. *Obs* plots
- geometric mean fold error (GMFE):

$$10^{\frac{\sum |\log(\frac{Pred}{Obs})|}{n}}$$

- number of AUCR and CMAXR falling within 2-fold error range and within the limits suggested by [Guest et al. 2011](#)
- detailed table of results for each study

In the plots,

- the dotted lines denote 0.50–2.00 (2-fold) criterion,
- the solid lines denote the limits suggested by [Guest et al. 2011](#),
- the bold solid line denotes the unity line,
- each color represents one combination of perpetrator and victim (parent drug or metabolite)

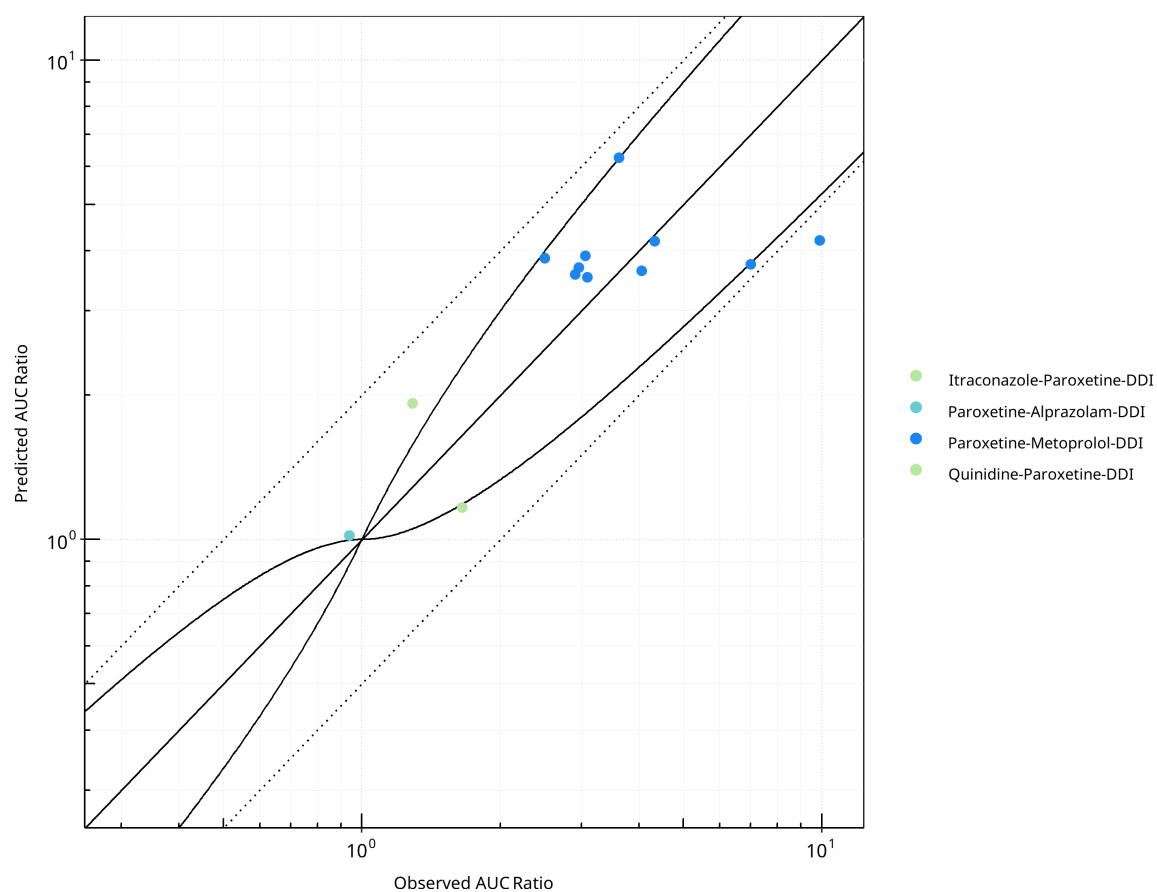


Figure 2-1: CYP2D6 DDIs (paroxetine). Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

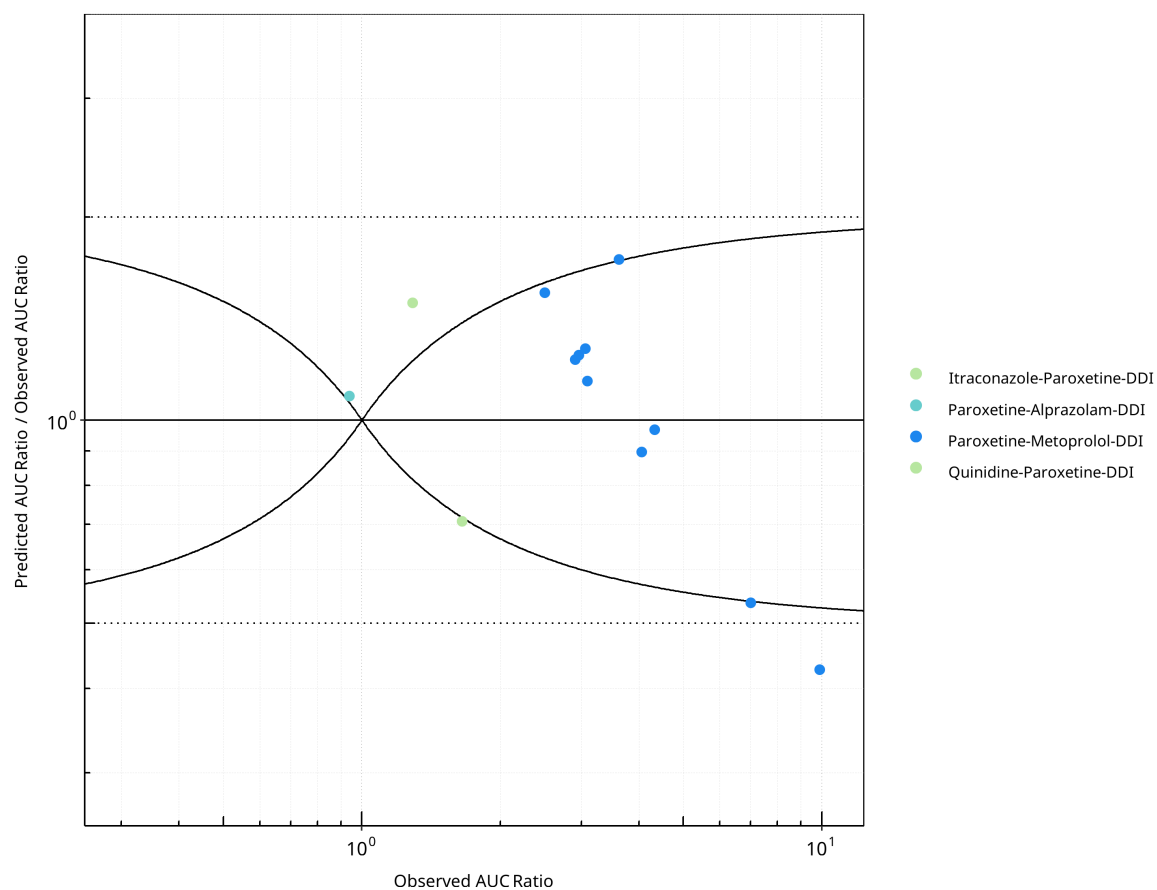


Figure 2-2: CYP2D6 DDIs (paroxetine). Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

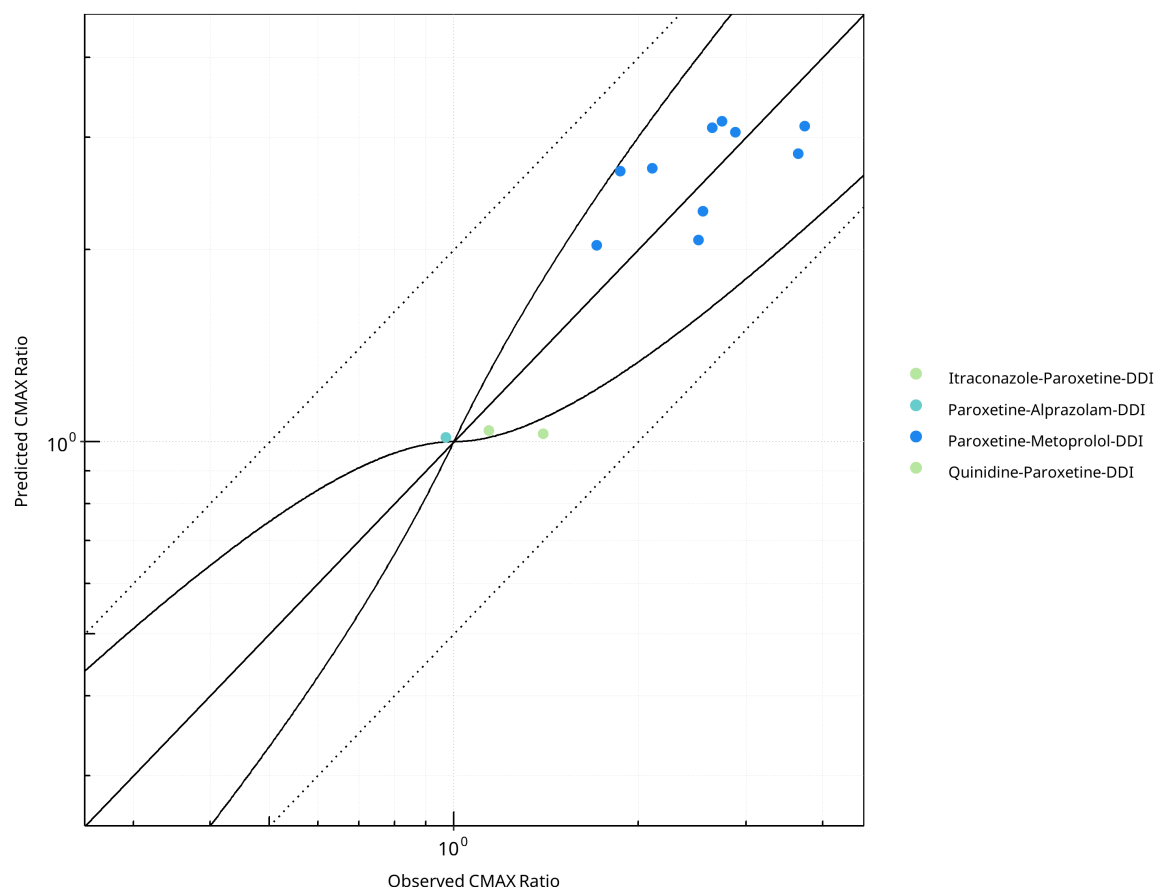


Figure 2-3: CYP2D6 DDIs (paroxetine). Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

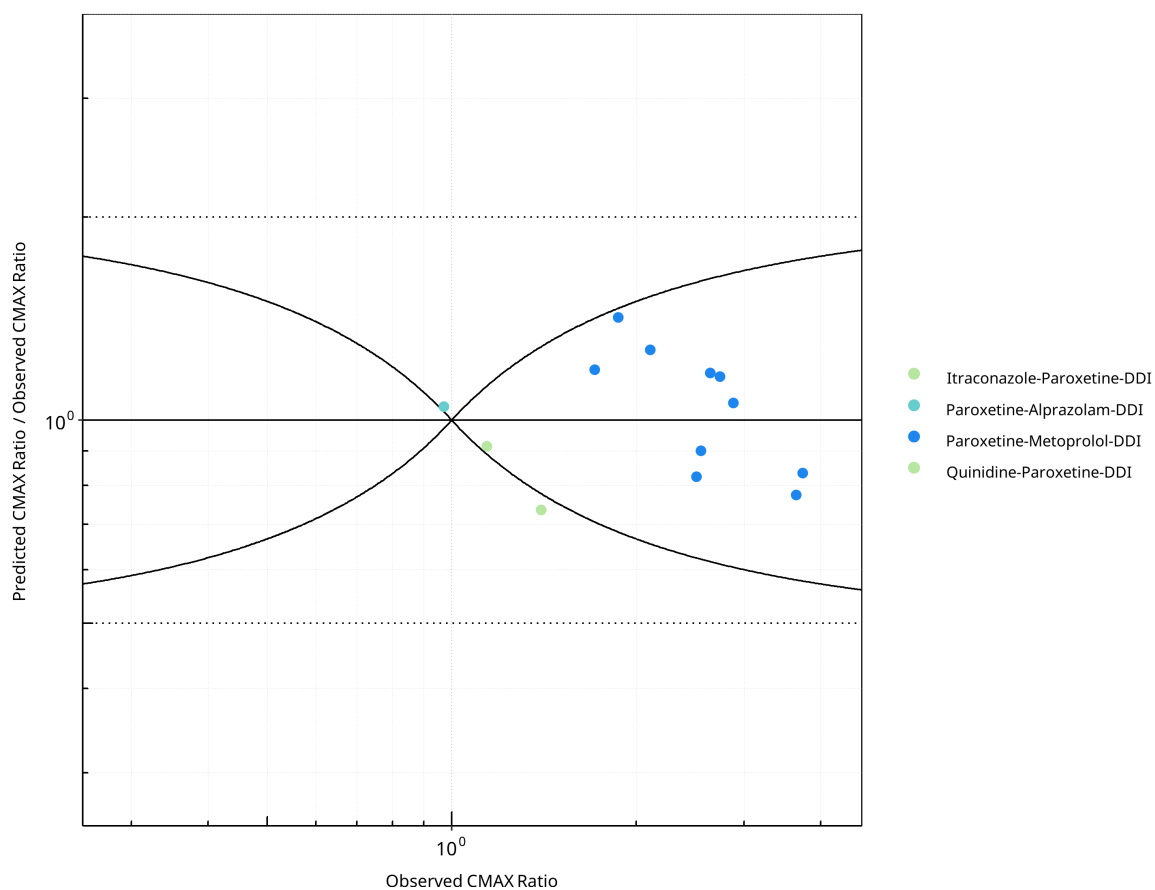


Figure 2-4: CYP2D6 DDIs (paroxetine). Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 2-1: GMFE for CYP2D6 DDIs (paroxetine) Ratio

PK parameter	GMFE
AUC	1.39
CMAX	1.19

Table 2-2: Summary table for CYP2D6 DDIs (paroxetine) - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	13	-
Points within Guest <i>et al.</i>	7	53.85
Points within 2 fold	12	92.31

Table 2-3: Summary table for CYP2D6 DDIs (paroxetine) - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

CMAX	Number	Ratio [%]
Points total	13	-
Points within Guest <i>et al.</i>	11	84.62
Points within 2 fold	13	100.00

Table 2-4: Summary table for CYP2D6 DDIs (paroxetine)

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
16954	Quinidine, 30 mg, PO, BID	Paroxetine, PO	1.93	1.29	1.50	1.04	1.14	0.92	Schoedel 2012
16980	Paroxetine, 20 mg, PO, QD	Alprazolam, PO	1.02	0.94	1.09	1.02	0.97	1.05	Calvo 2004
17116	Paroxetine, 10.10.2010 mg, PO, QD/BID	Metoprolol, PO	4.20	4.33	0.97	2.07	2.51	0.83	Stout 2011
17118	Paroxetine, 10.10.2010 mg, PO, QD/BID	Metoprolol, PO	6.28	3.62	1.73	2.84	3.65	0.78	Stout 2011
17120	Paroxetine, 20 mg, PO, QD	R- Metoprolol, PO	3.70	2.96	1.25	3.18	2.74	1.16	Parker 2011
17122	Paroxetine, 20 mg, PO, QD	S- Metoprolol, PO	3.59	2.91	1.23	3.11	2.64	1.18	Parker 2011
17124	Paroxetine, 20 mg, PO, QD	R- Metoprolol, PO	3.92	3.06	1.28	2.69	2.11	1.27	Parker 2011
17126	Paroxetine, 20 mg, PO, QD	S- Metoprolol, PO	3.87	2.50	1.55	2.66	1.87	1.42	Parker 2011
17128	Paroxetine, 20 mg, PO, QD	R- Metoprolol, PO	3.65	4.06	0.90	3.13	3.74	0.84	Parker 2011
17130	Paroxetine, 20 mg, PO, QD	S- Metoprolol, PO	3.54	3.09	1.14	3.06	2.88	1.06	Parker 2011
17132	Paroxetine, 10 mg, PO, BID	R- Metoprolol, PO	4.22	9.88	0.43	2.30	2.55	0.90	Hemeryck 2000
17134	Paroxetine, 10 mg, PO, BID	S- Metoprolol, PO	3.77	7.01	0.54	2.04	1.71	1.19	Hemeryck 2000
17158	Paroxetine, 100 mg, PO, BID	Paroxetine, PO	1.17	1.65	0.71	1.03	1.40	0.74	Yasui- Furukori 2007

2.1 DDI Network Evaluations

Figure 1 shows the developed CYP2D6 DDGI network of interacting perpetrator and victim drugs. (a) Schematic illustration of the modeled interactions of CYP2D6 perpetrator and victim drugs. Black arrows indicate metabolism or

transport, green arrows indicate induction, red solid lines indicate competitive inhibition, red dashed lines down-regulation (bupropion), noncompetitive inhibition (verapamil P-gp inhibition) or mechanism-based inactivation (others). (b-c) Drug-drug-(gene) interaction matrix for modeled interactions mediated by (a) CYP2D6 and (b) CYP3A4 and P-gp. Colors indicate categories according to the FDA's Examples of Drugs that Interact with CYP Enzymes and Transporter Systems.⁴² Height of the grey ribbons indicates the number of clinical studies for the respective interaction covered by the network, numbers in brackets indicate the number of clinical interaction studies for the corresponding compound. CYP: cytochrome P450, P-gp: P-glycoprotein.

Figure 1: CYP2D6 DDGI modeling network

The following victim and/or perpetrator PBPK models were applied:

- **Alprazolam** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Alprazolam-Model/blob/master/Alprazolam-Model.json>
- **Atomoxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Atomoxetine-Model/blob/main/atomoxetine-model.json>
- **Bupropion** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Bupropion-Model/blob/main/Bupropion-Model.json>
- **Carbamazepine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Carbamazepine-Model/blob/main/Carbamazepine-Model.json>
- **Cimetidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Cimetidine-Model/blob/master/Cimetidine-Model.json>
- **Clarithromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Clarithromycin-Model/blob/master/Clarithromycin-Model.json>
- **(E)-Clomiphene** Model snapshot: [https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/\(E\)-clomiphene-DGI-Model.json](https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/(E)-clomiphene-DGI-Model.json)
- **Desipramine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Desipramine-Model/blob/main/Desipramine-Model.json>
- **Dextromethorphan** Model snapshot: https://github.com/Open-Systems-Pharmacology/Dextromethorphan-Model/blob/main/dextromethorphan_aggregated_simulations.json
- **Digoxin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Digoxin-Model/blob/master/Digoxin.json>
- **Erythromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Erythromycin-Model/blob/master/Erythromycin-Model.json>
- **Fluvoxamine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Fluvoxamine-Model/blob/master/Fluvoxamine-Model.json>
- **Itraconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Itraconazole-Model/blob/master/Itraconazole-Model.json>
- **Ketoconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Ketoconazole-Model/blob/main/Ketoconazole-Model.json>
- **Metoprolol** Model file: <https://github.com/Open-Systems-Pharmacology/Metoprolol-Model/blob/main/Metoprolol-Model.pksim5>
- **Mexiletine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Mexiletine-Model/blob/main/Mexiletine-Model.json>
- **Omeprazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Omeprazole-Model/blob/main/Omeprazole-Model.json>
- **Paroxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Paroxetine-Model/blob/main/paroxetine-model.json>

- **Quinidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Quinidine-Model/blob/main/Quinidine-Model.json>
- **Rifampicin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Rifampicin-Model/blob/master/Rifampicin-Model.json>
- **Risperidone** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Risperidone-Model/blob/main/risperidone-model.json>
- **Verapamil** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Verapamil-Model/blob/master/Verapamil-Model.json>

The following interaction scenarios were predicted and used to qualify the final DDGI network:

- Alprazolam as victim:
 - Paroxetine-alprazolam-DDI
- Atomoxetine as victim:
 - Bupropion-atomoxetine-DDGI
 - Fluvoxamine-atomoxetine-DDI
 - Paroxetine-atomoxetine-DDGI
- (E)-Clomiphene as victim:
 - Clarithromycin-clomiphene-DDGI
 - Paroxetine-clomiphene-DDGI
- Desipramine as victim:
 - Atomoxetine-desipramine-DDI
 - Bupropion-desipramine-DDI
 - Paroxetine-desipramine-DDGI
 - Quinidine-desipramine-DDI
- Dextromethorphan as victim:
 - Paroxetine-dextromethorphan-DDGI
 - Quinidine-dextromethorphan-DDI
- Digoxin as victim:
 - Quinidine-digoxin-DDI
- Metoprolol as victim:
 - Cimetidine-metoprolol-DDI
 - Paroxetine-metoprolol-DDI
 - Quinidine-metoprolol-DDGI
 - Rifampicin-metoprolol-DDI
- Mexiletine as victim:
 - Quinidine-mexiletine-DDGI
- Midazolam as victim:
 - Atomoxetine-midazolam-DDI
- Paroxetine as victim:
 - Itraconazole-paroxetine-DDI
 - Quinidine-paroxetine-DDI
- Quinidine as victim:
 - Carbamazepine-quinidine-DDI
 - Cimetidine-quinidine-DDI

- Erythromycin-quinidine-DDI
- Fluvoxamine-quinidine-DDI
- Itraconazole-quinidine-DDI
- Omeprazole-quinidine-DDI
- Rifampicin-quinidine-DDI
- Verapamil-quinidine-DDI
- Risperidone as victim:
 - Ketoconazole-risperidone-DDI
 - Rifampicin-risperidone-DDI
 - Verapamil-risperidone-DDI

The published DD(G)I studies between the respective perpetrators and victim drugs were simulated and compared to observed data. The following sections give an overview of the clinical studies being part of this qualification report.

2.1.1 Paroxetine - Alprazolam DDI

The paroxetine-alprazolam interaction was evaluated using clinical DDI studies listed in [Table 1](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Calvo 2004	po	1 mg q.d.	European	f	22	+/- PAR, 20 mg q.d. po

Table 1: f: female, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.2 Paroxetine - Metoprolol DDI

The paroxetine-metoprolol interaction was evaluated using clinical DDI studies listed in [Table 15](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Stout 2011	po	50 mg s.d.	American	m	10	+/- PAR, 10 mg b.i.d. po
Stout 2011	po	100 mg s.d.	American	m	10	+/- PAR, 10 mg b.i.d. po
Parker 2011	po	100 mg s.d.	American	m	12	+/- PAR, 20 mg q.d. po
Parker 2011	po	100 mg b.i.d.	American	m	12	+/- PAR, 20 mg q.d. po
Parker 2011	po	200 mg s.d.	American	m	12	+/- PAR, 20 mg q.d. po
Hemeryck 2000	po	100 mg s.d.	European	m	8	+/- PAR, 10 mg b.i.d. po

Table 15: b.i.d.: twice daily, m: male, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.3 Itraconazole - Paroxetine DDI

The itraconazole-paroxetine interaction was evaluated using clinical DDI studies listed in [Table 20](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Yasui-Furukori 2007	po	20 mg s.d.	Japanese	f	13	+/- ITR, 100 mg b.i.d. po

Table 20: b.i.d.: twice daily, f: female, ITR: itraconazole, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.1.4 Quinidine - Paroxetine DDI

The quinidine-paroxetine interaction was evaluated using clinical DDI studies listed in [Table 21](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Schoedel 2012	po	20 mg q.d.	American	m	14 NM	+/- QUI, 30 mg b.i.d. po

Table 21: m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, po: oral, pop.: population used in simulations, q.d.: once daily, QUI: quinidine. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2.2 DDI Concentration-Time Profiles

The following section shows concentration-time profiles of the victim drugs of the simulated DD(G)I studies in comparison to observed data.

2.2.1 Paroxetine - Alprazolam DDI

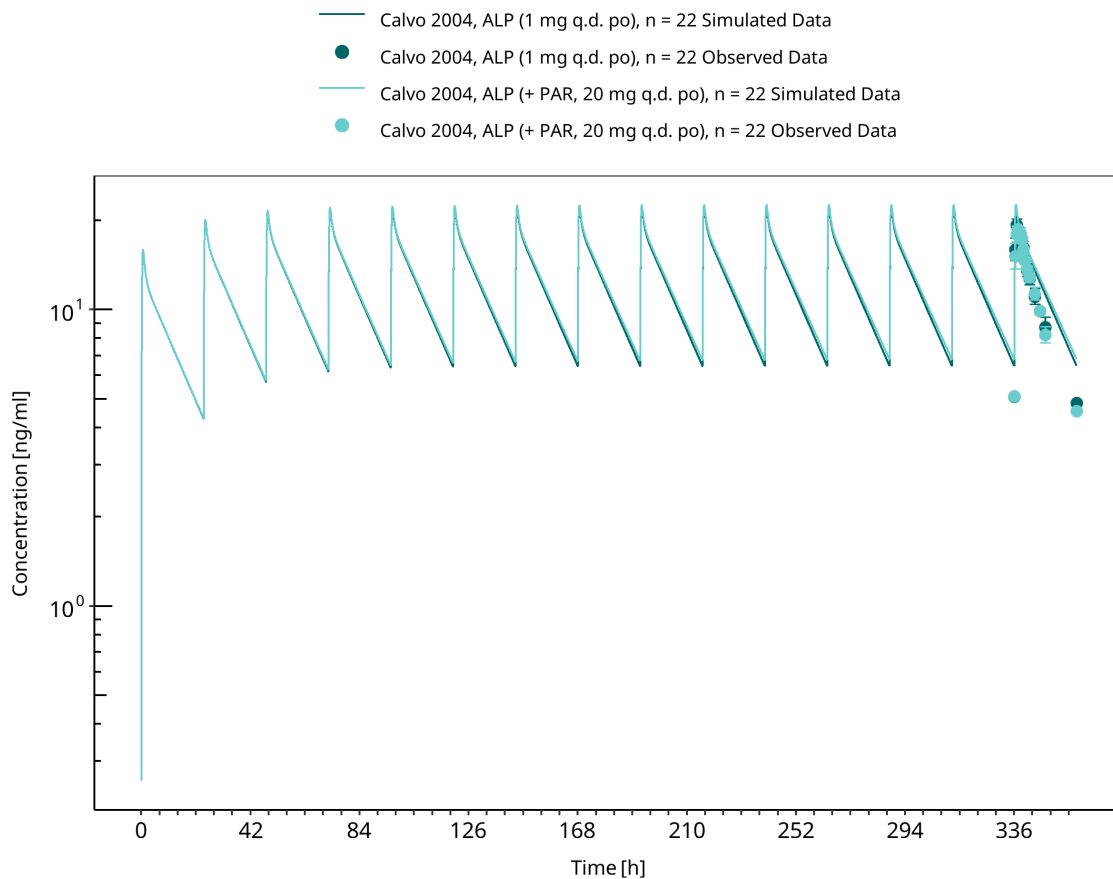


Figure 2-6: Paroxetine - Alprazolam DDI

2.2.2 Paroxetine - Metoprolol DDI

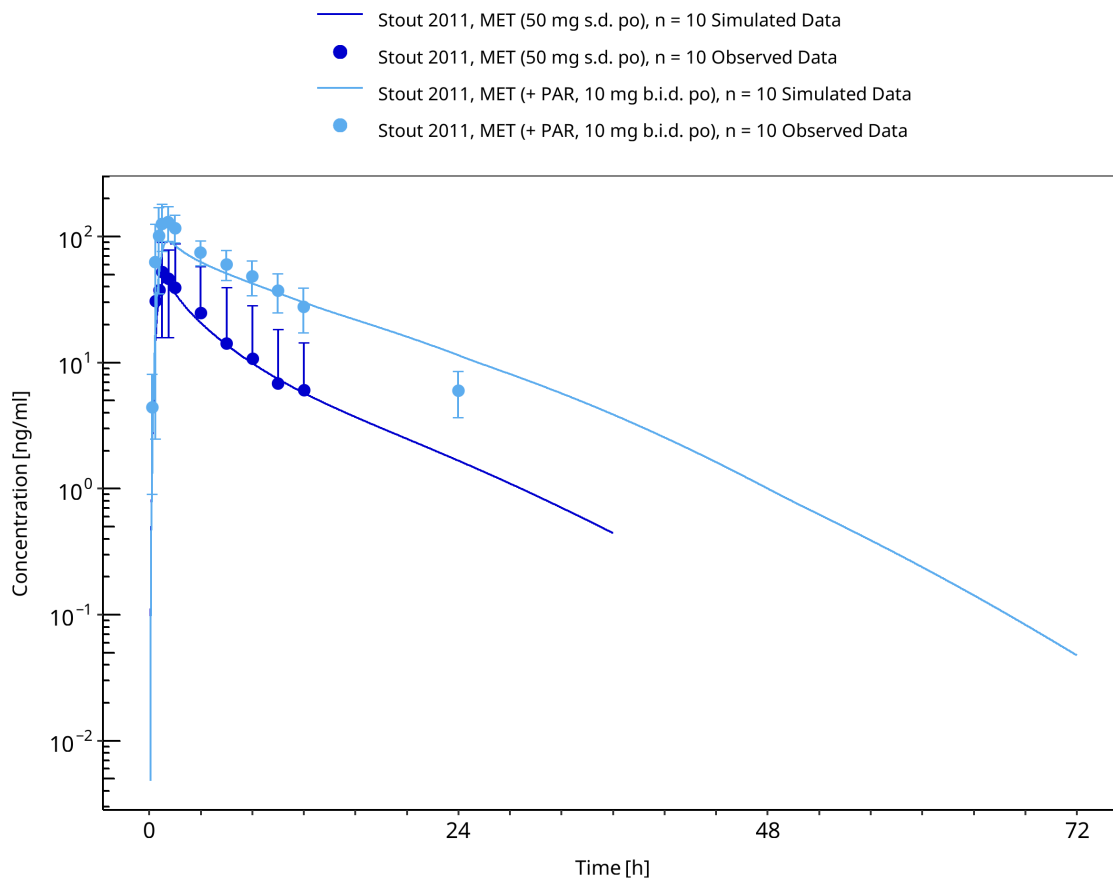


Figure 2-7: Paroxetine - Metoprolol DDI

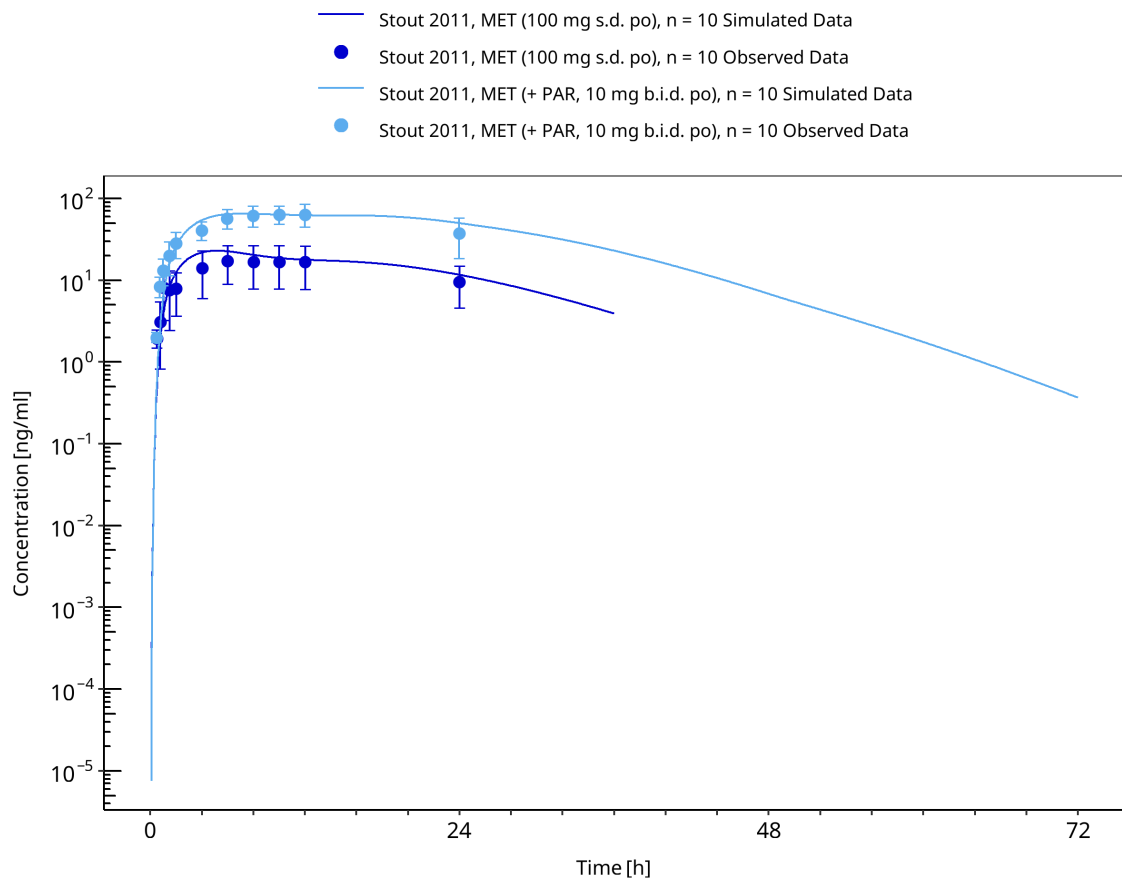


Figure 2-8: Paroxetine - Metoprolol DDI

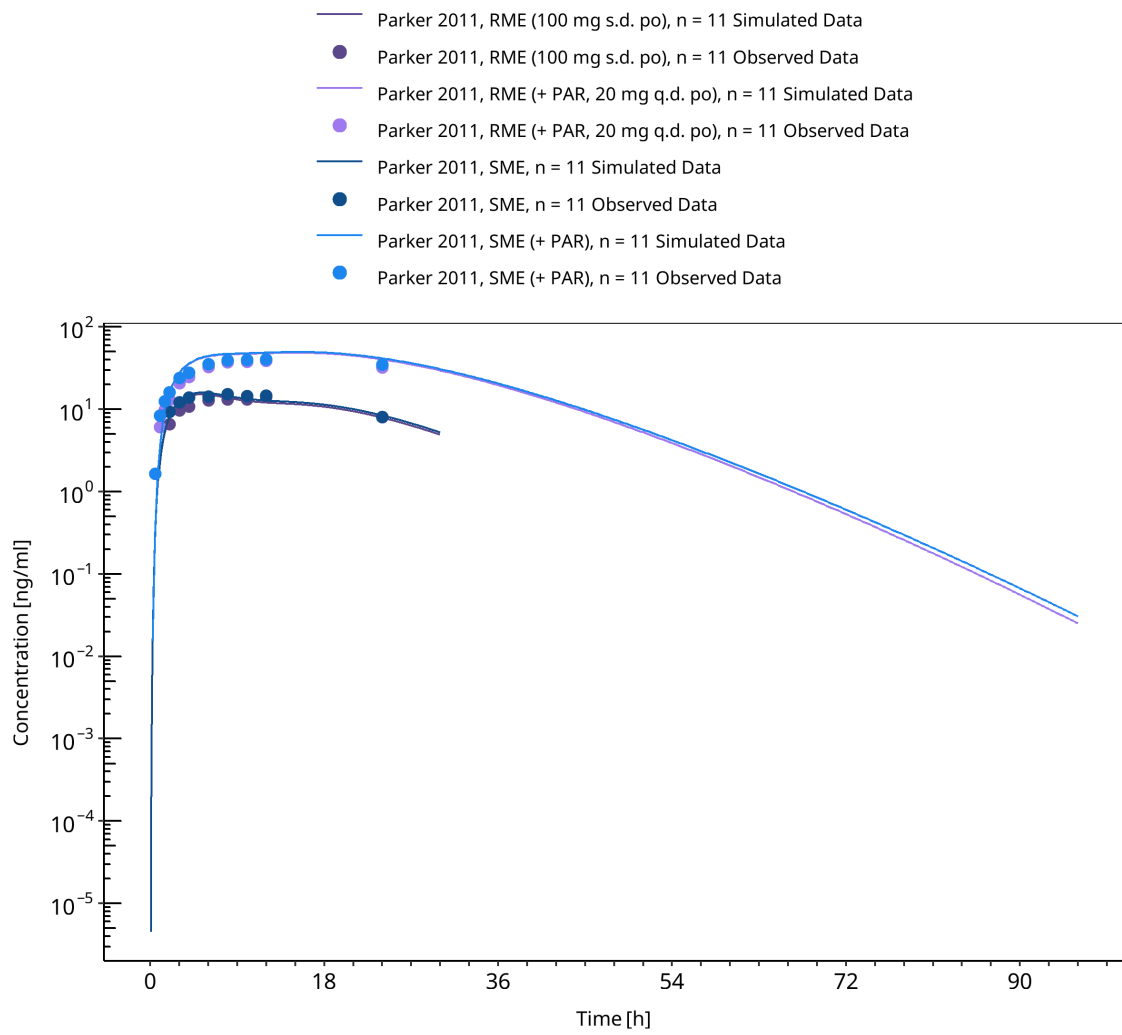


Figure 2-9: Paroxetine - Metoprolol DDI

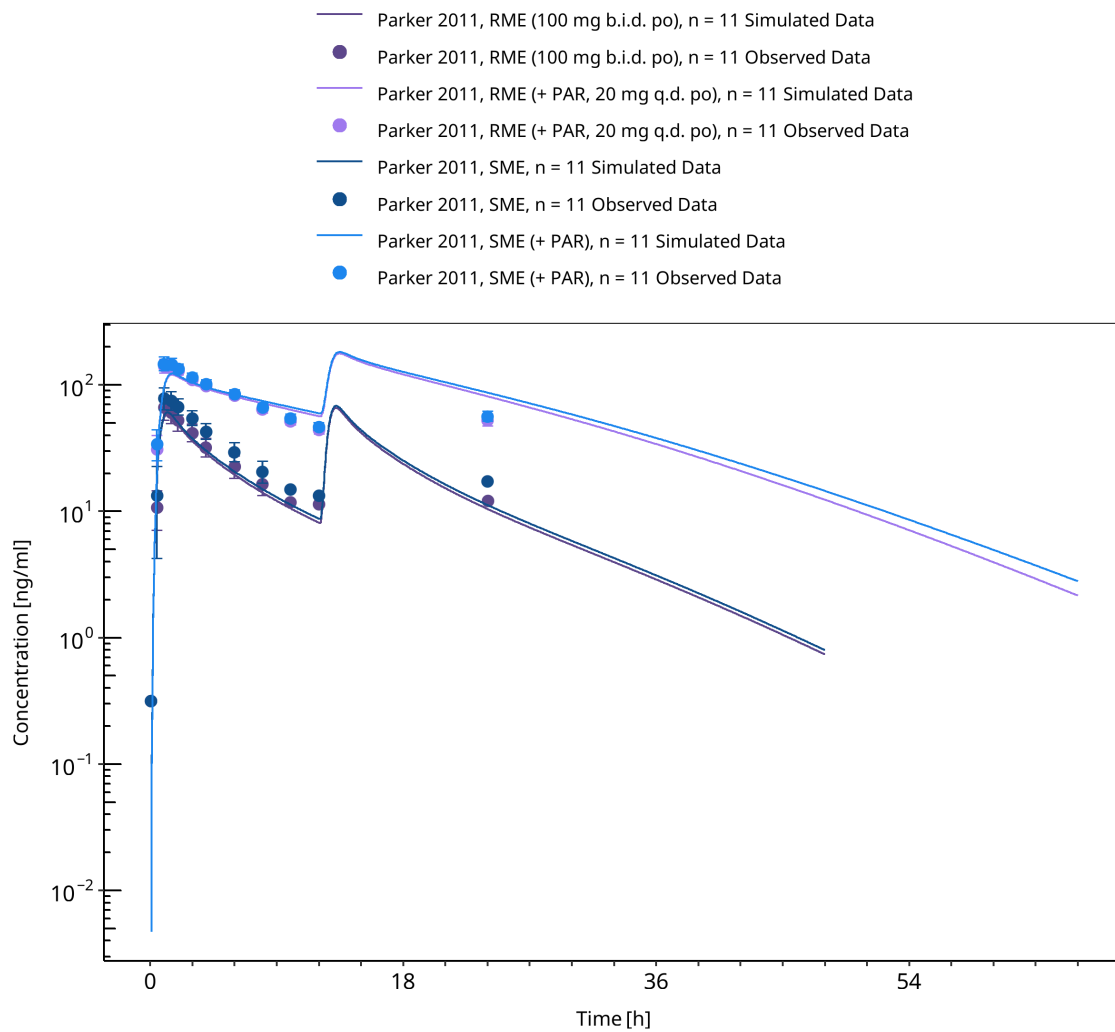


Figure 2-10: Paroxetine - Metoprolol DDI

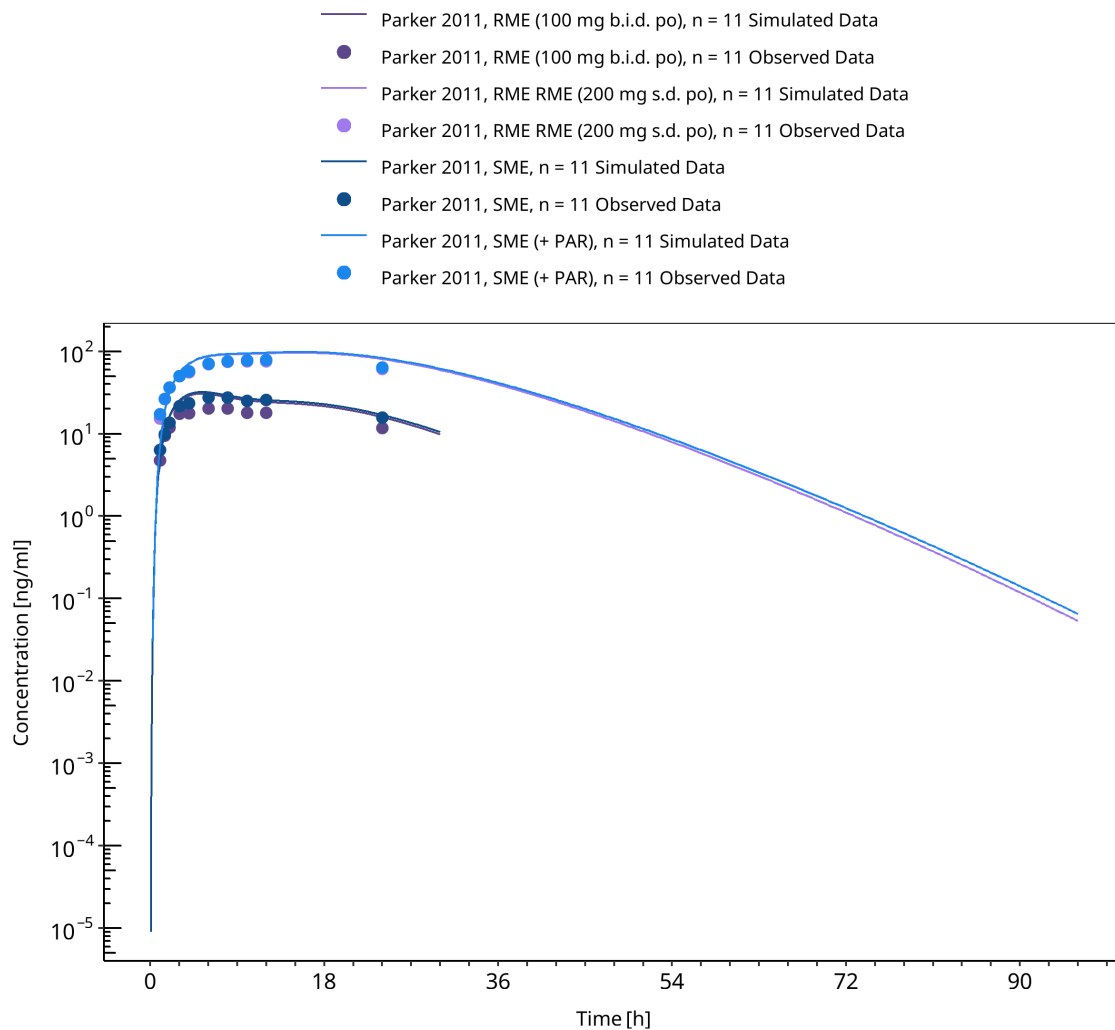


Figure 2-11: Paroxetine - Metoprolol DDI

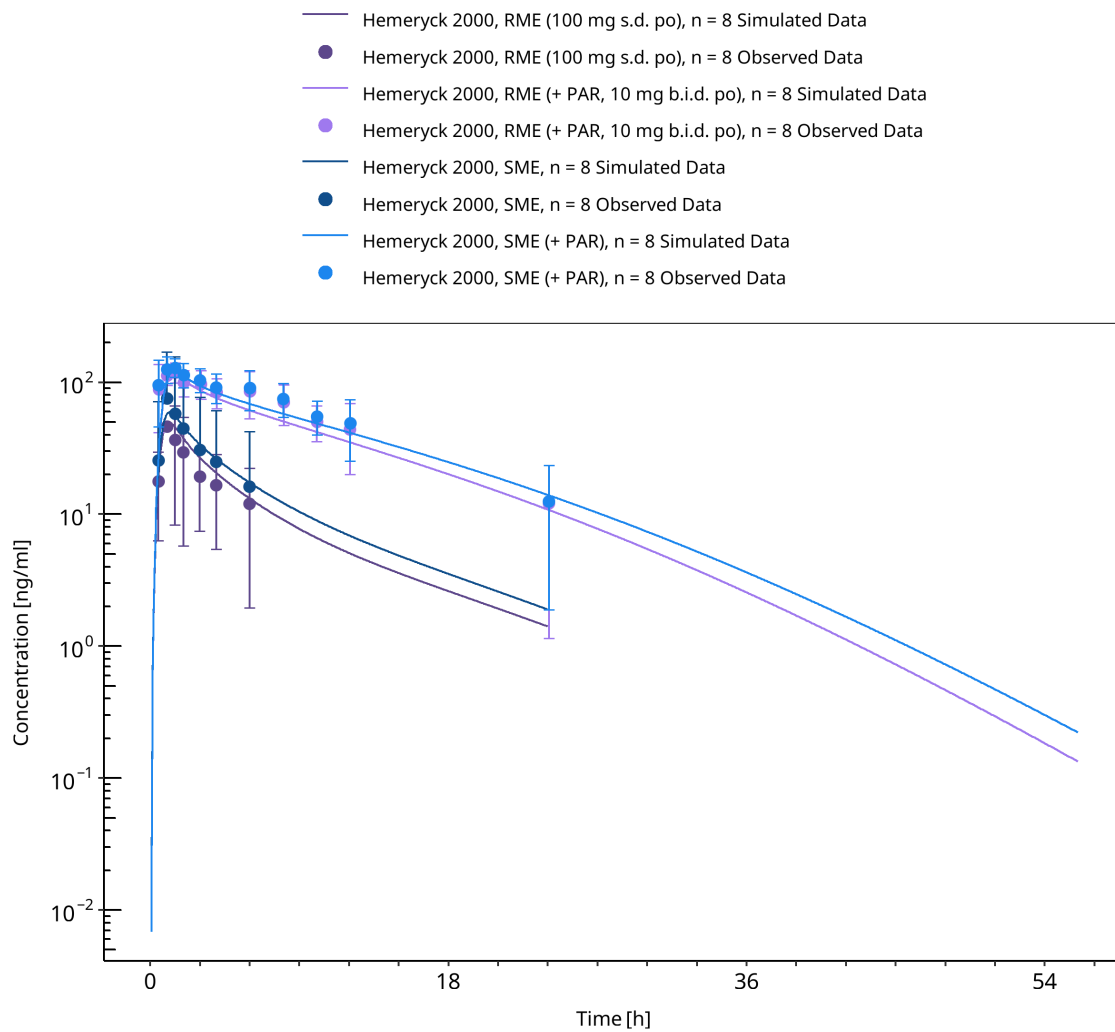


Figure 2-12: Paroxetine - Metoprolol DDI

2.2.3 Itraconazole - Paroxetine DDI

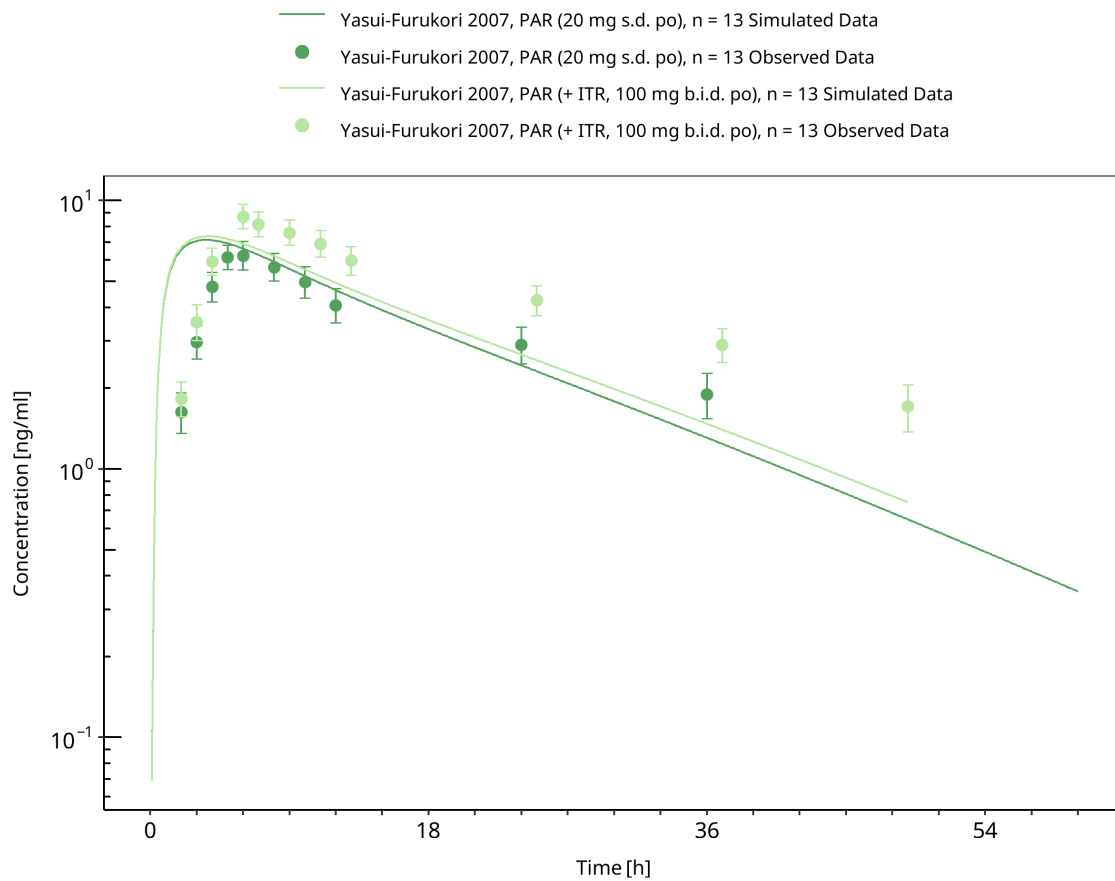


Figure 2-13: Itraconazole - Paroxetine DDI

2.2.4 Quinidine - Paroxetine DDI

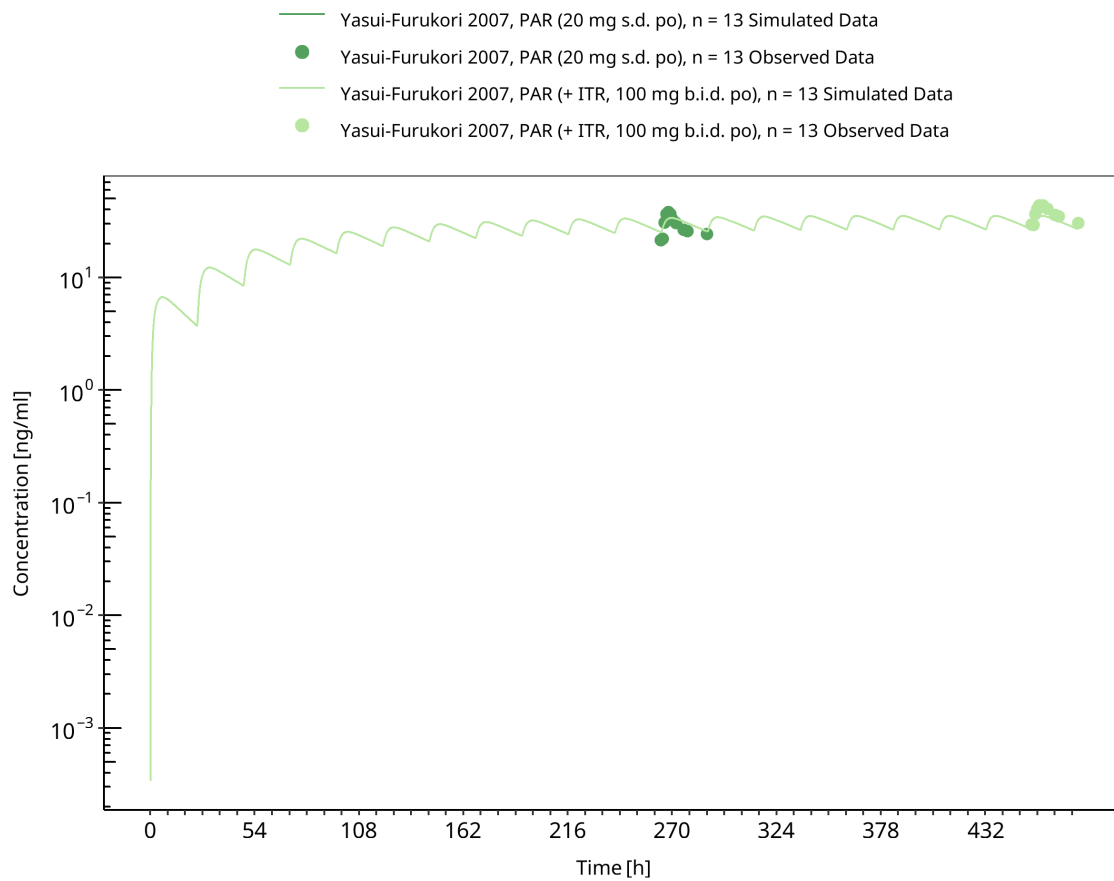


Figure 2-14: Quinidine - Paroxetine DDI

3 DDGI Evaluations

The following section shows the correlations between observed and model-predicted AUC_{last} and C_{max} ratios, respectively.

Specifically, the PBPK model performance for the PK parameters **AUC ratio (AUCR)** and **C_{max} ratio (CmaxR)** is assessed via:

- predicted (*Pred*) vs. observed (*Obs*) plots
- *Pred/Obs* vs. *Obs* plots
- geometric mean fold error (GMFE):

$$10^{\frac{\sum |\log(\frac{Pred}{Obs})|}{n}}$$

- number of AUCR and CMAXR falling within 2-fold error range and within the limits suggested by [Guest et al. 2011](#)
- detailed table of results for each study

In the plots,

- the dotted lines denote 0.50–2.00 (2-fold) criterion,
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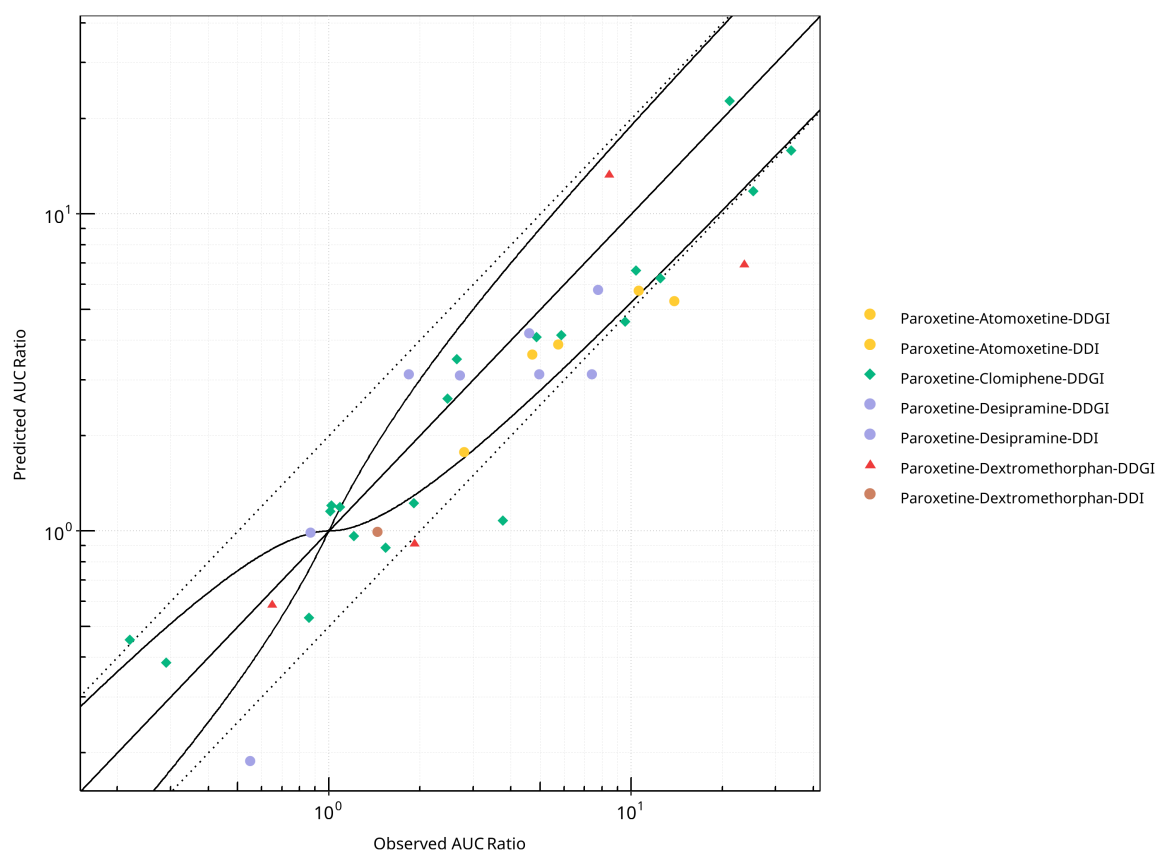


Figure 3-1: CYP2D6 DDGIs (paroxetine). Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

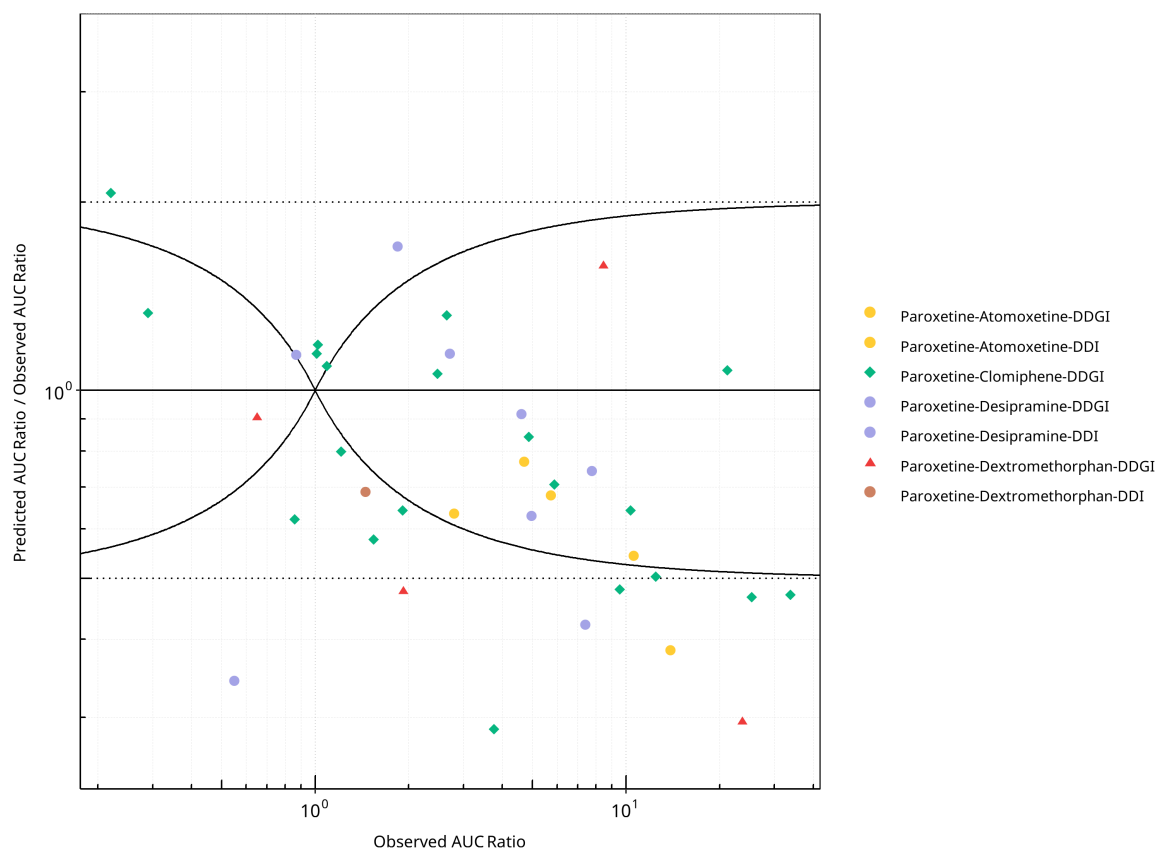


Figure 3-2: CYP2D6 DDGIs (paroxetine). Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

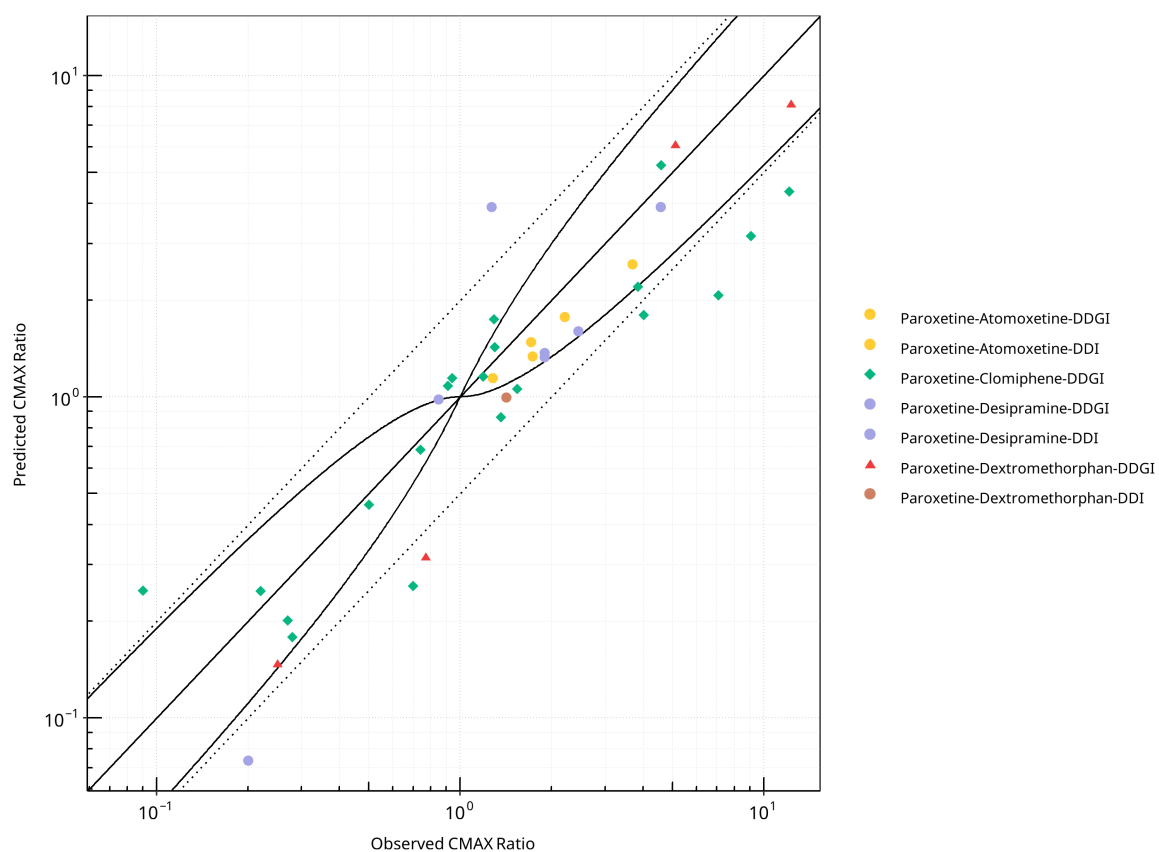


Figure 3-3: CYP2D6 DDGIs (paroxetine). Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

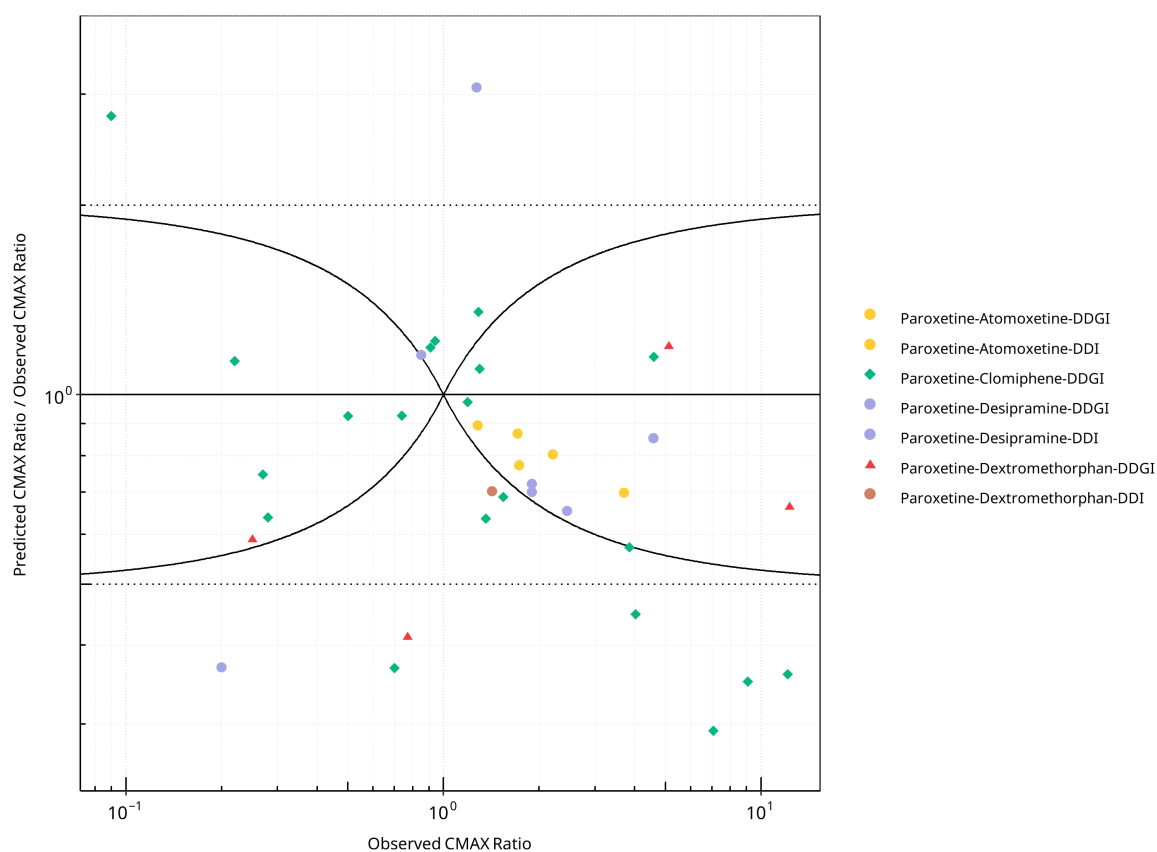


Figure 3-4: CYP2D6 DDGIs (paroxetine). Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 3-1: GMFE for CYP2D6 DDGIs (paroxetine) Ratio

PK parameter	GMFE
AUC	1.60
CMAX	1.55

Table 3-2: Summary table for CYP2D6 DDGIs (paroxetine) - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	38	-
Points within Guest <i>et al.</i>	17	44.74
Points within 2 fold	28	73.68

Table 3-3: Summary table for CYP2D6 DDGs (paroxetine) - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

CMAX	Number	Ratio [%]
Points total	38	-
Points within Guest <i>et al.</i>	21	55.26
Points within 2 fold	29	76.32

Table 3-4: Summary table for CYP2D6 DDGs (paroxetine)

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
16988	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	1.78	2.80	0.64	1.15	1.28	0.90	Jung 2020
16990	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	3.62	4.70	0.77	1.49	1.71	0.87	Jung 2020
16992	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	5.76	10.57	0.54	1.78	2.21	0.81	Jung 2020
16994	Paroxetine, 20/20 mg, PO, BID/QD	Atomoxetine, PO	3.90	5.73	0.68	1.34	1.73	0.77	Todor 2015
16996	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	5.34	13.87	0.38	2.59	3.70	0.70	Belle 2002
17038	Paroxetine, 40 mg, PO, QD	(E)-Clomiphen, PO	1.19	1.09	1.10	1.08	0.91	1.19	Mürdter 2016
17040	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophen, PO	1.21	1.02	1.19	1.16	1.19	0.98	Mürdter 2016
17042	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphen, PO	0.97	1.21	0.80	0.87	1.36	0.64	Mürdter 2016
17044	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphen, PO	1.16	1.01	1.15	1.15	0.94	1.22	Mürdter 2016
17046	Paroxetine, 40 mg, PO, QD	(E)-Clomiphen, PO	2.63	2.47	1.06	1.43	1.30	1.10	Mürdter 2016
17048	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophen, PO	0.46	0.22	2.07	0.25	0.09	2.78	Mürdter 2016
17050	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphen, PO	4.10	4.86	0.84	1.75	1.29	1.36	Mürdter 2016
17052	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphen, PO	0.39	0.29	1.33	0.25	0.22	1.13	Mürdter 2016
17054	Paroxetine, 40 mg, PO, QD	(E)-Clomiphen, PO	4.59	9.53	0.48	1.81	4.02	0.45	Mürdter 2016

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17056	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	0.54	0.86	0.62	0.18	0.28	0.64	Mürdter 2016
17058	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	11.85	25.32	0.47	3.18	9.07	0.35	Mürdter 2016
17060	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	1.23	1.91	0.64	0.46	0.50	0.93	Mürdter 2016
17062	Paroxetine, 40 mg, PO, QD	(E)-Clomiphene, PO	6.66	10.35	0.64	2.21	3.85	0.57	Mürdter 2016
17064	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	0.89	1.54	0.58	0.20	0.27	0.75	Mürdter 2016
17066	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	22.80	21.14	1.08	5.28	4.59	1.15	Mürdter 2016
17068	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	3.50	2.65	1.32	0.69	0.74	0.93	Mürdter 2016
17070	Paroxetine, 40 mg, PO, QD	(E)-Clomiphene, PO	6.29	12.47	0.50	2.08	7.08	0.29	Mürdter 2016
17072	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	1.08	3.76	0.29	0.26	0.70	0.37	Mürdter 2016
17074	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	15.92	33.72	0.47	4.38	12.14	0.36	Mürdter 2016
17076	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	4.16	5.87	0.71	1.06	1.54	0.69	Mürdter 2016
17082	Paroxetine, 20 mg, PO, QD	Desipramine, PO	3.11	2.71	1.15	1.37	1.90	0.72	Nichols 2009
17084	Paroxetine, 20 mg, PO, QD	2- Hydroxydesipramine, PO	0.19	0.55	0.34	0.07	0.20	0.37	Nichols 2009
17086	Paroxetine, 20 mg, PO, QD	Desipramine, PO	0.99	0.87	1.14	0.99	0.85	1.16	Brøsen 1993

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17088	Paroxetine, 20 mg, PO, QD	Desipramine, PO	4.22	4.60	0.92	1.33	1.90	0.70	Brøsen 1993
17090	Paroxetine, 20 mg, PO, QD	Desipramine, PO	5.78	7.76	0.74	1.61	2.45	0.66	Brøsen 1993
17092	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	1.84	1.70	3.91	1.27	3.08	Aldermann 1997
17093	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	4.96	0.63	3.91	4.58	0.85	Aldermann 1997
17094	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	7.40	0.42	3.91	4.58	0.85	Aldermann 1997
17098	Paroxetine, 20 mg, PO,	Dextromethorphan, PO	13.46	8.45	1.59	6.14	5.11	1.20	Storelli 2018
17100	Paroxetine, 20 mg, PO,	Dextrophan, PO	0.59	0.65	0.91	0.15	0.25	0.59	Storelli 2018
17102	Paroxetine, 20 mg, PO,	Dextromethorphan, PO	7.01	23.61	0.30	8.21	12.29	0.67	Storelli 2018
17104	Paroxetine, 20 mg, PO,	Dextrophan, PO	0.92	1.92	0.48	0.32	0.77	0.42	Storelli 2018
17106	Paroxetine, 20 mg, PO, QD	Dextromethorphan, PO	1.00	1.45	0.69	1.00	1.42	0.70	Schoedel 2012

3.1 DDGI Network Evaluations

Figure 1 shows the developed CYP2D6 DDGI network of interacting perpetrator and victim drugs. (a) Schematic illustration of the modeled interactions of CYP2D6 perpetrator and victim drugs. Black arrows indicate metabolism or transport, green arrows indicate induction, red solid lines indicate competitive inhibition, red dashed lines down-regulation (bupropion), noncompetitive inhibition (verapamil P-gp inhibition) or mechanism-based inactivation (others). (b-c) Drug-drug-(gene) interaction matrix for modeled interactions mediated by (a) CYP2D6 and (b) CYP3A4 and P-gp. Colors indicate categories according to the FDA's Examples of Drugs that Interact with CYP Enzymes and Transporter Systems.⁴² Height of the grey ribbons indicates the number of clinical studies for the respective interaction covered by the network, numbers in brackets indicate the number of clinical interaction studies for the corresponding compound. CYP: cytochrome P450, P-gp: P-glycoprotein.

Figure 1: CYP2D6 DDGI modeling network

The following victim and/or perpetrator PBPK models were applied:

- **Alprazolam** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Alprazolam-Model/blob/master/Alprazolam-Model.json>
- **Atomoxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Atomoxetine-Model/blob/main/atomoxetine-model.json>
- **Bupropion** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Bupropion-Model/blob/main/Bupropion-Model.json>
- **Carbamazepine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Carbamazepine-Model/blob/main/Carbamazepine-Model.json>
- **Cimetidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Cimetidine-Model/blob/master/Cimetidine-Model.json>
- **Clarithromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Clarithromycin-Model/blob/master/Clarithromycin-Model.json>
- **(E)-Clomiphene** Model snapshot: [https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/\(E\)-clomiphene-DGI-Model.json](https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/(E)-clomiphene-DGI-Model.json)
- **Desipramine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Desipramine-Model/blob/main/Desipramine-Model.json>
- **Dextromethorphan** Model snapshot: https://github.com/Open-Systems-Pharmacology/Dextromethorphan-Model/blob/main/dextromethorphan_aggregated_simulations.json
- **Digoxin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Digoxin-Model/blob/master/Digoxin.json>
- **Erythromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Erythromycin-Model/blob/master/Erythromycin-Model.json>
- **Fluvoxamine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Fluvoxamine-Model/blob/master/Fluvoxamine-Model.json>
- **Itraconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Itraconazole-Model/blob/master/Itraconazole-Model.json>
- **Ketoconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Ketoconazole-Model/blob/main/Ketoconazole-Model.json>
- **Metoprolol** Model file: <https://github.com/Open-Systems-Pharmacology/Metoprolol-Model/blob/main/Metoprolol-Model.pksim5>
- **Mexiletine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Mexiletine-Model/blob/main/Mexiletine-Model.json>
- **Omeprazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Omeprazole-Model/blob/main/Omeprazole-Model.json>
- **Paroxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Paroxetine-Model/blob/main/paroxetine-model.json>
- **Quinidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Quinidine-Model/blob/main/Quinidine-Model.json>
- **Rifampicin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Rifampicin-Model/blob/master/Rifampicin-Model.json>
- **Risperidone** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Risperidone-Model/blob/main/risperidone-model.json>
- **Verapamil** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Verapamil-Model/blob/master/Verapamil-Model.json>

The following interaction scenarios were predicted and used to qualify the final DDGI network:

- Alprazolam as victim:
 - Paroxetine-alprazolam-DDI
- Atomoxetine as victim:
 - Bupropion-atomoxetine-DDGI
 - Fluvoxamine-atomoxetine-DDI
 - Paroxetine-atomoxetine-DDGI
- (E)-Clomiphene as victim:
 - Clarithromycin-clomiphene-DDGI
 - Paroxetine-clomiphene-DDGI
- Desipramine as victim:
 - Atomoxetine-desipramine-DDI
 - Bupropion-desipramine-DDI
 - Paroxetine-desipramine-DDGI
 - Quinidine-desipramine-DDI
- Dextromethorphan as victim:
 - Paroxetine-dextromethorphan-DDGI
 - Quinidine-dextromethorphan-DDI
- Digoxin as victim:
 - Quinidine-digoxin-DDI
- Metoprolol as victim:
 - Cimetidine-metoprolol-DDI
 - Paroxetine-metoprolol-DDI
 - Quinidine-metoprolol-DDGI
 - Rifampicin-metoprolol-DDI
- Mexiletine as victim:
 - Quinidine-mexiletine-DDGI
- Midazolam as victim:
 - Atomoxetine-midazolam-DDI
- Paroxetine as victim:
 - Itraconazole-paroxetine-DDI
 - Quinidine-paroxetine-DDI
- Quinidine as victim:
 - Carbamazepine-quinidine-DDI
 - Cimetidine-quinidine-DDI
 - Erythromycin-quinidine-DDI
 - Fluvoxamine-quinidine-DDI
 - Itraconazole-quinidine-DDI
 - Omeprazole-quinidine-DDI
 - Rifampicin-quinidine-DDI
 - Verapamil-quinidine-DDI
- Risperidone as victim:
 - Ketoconazole-risperidone-DDI
 - Rifampicin-risperidone-DDI

- Verapamil-risperidone-DDI

The published DD(G)I studies between the respective perpetrators and victim drugs were simulated and compared to observed data. The following sections give an overview of the clinical studies being part of this qualification report.

3.1.1 Paroxetine - Atomoxetine DDGI

The paroxetine-atomoxetine interaction was evaluated using clinical DDGI studies listed in [Table 4](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Jung 2020	po	20 mg s.d.	Asian	m	7 AS0	+/- PAR, 20 mg q.d. po
Jung 2020	po	20 mg s.d.	Asian	m	9 AS1.25	+/- PAR, 20 mg q.d. po
Jung 2020	po	20 mg s.d.	Asian	m	10 AS2	+/- PAR, 20 mg q.d. po
Todor 2015	po	25 mg s.d.	European	m	22	+/- PAR, 20 mg b.i.d./q.d. po
Belle 2002	po	20 mg b.i.d.	American	m	22	+/- PAR, 20 mg q.d. po

Table 4: AS: CYP2D6 activity score, b.i.d.: twice daily, m: male, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

3.1.2 Paroxetine - Clomiphene DDGI

The paroxetine-clomiphene interaction was evaluated using clinical DDGI studies listed in [Table 6](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Mürdter 2016	po	42 mg s.d.	European	f	4 AS0	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	4 AS0.5	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS1	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS2	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS3	+/- PAR, 40 mg q.d. po

Table 6: AS: CYP2D6 activity score, b.i.d.: twice daily, f: female, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

3.1.3 Paroxetine - Desipramine DDGI

The paroxetine-desipramine interaction was evaluated using clinical DDGI studies listed in [Table 9](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Nichols 2009	po	50 mg s.d.	American	m	20	+/- PAR, 20 mg q.d. po
Brøsen 1993	po	100 mg s.d.	European	m	8 PM	+/- PAR, 20 mg q.d. po
Brøsen 1993	po	100 mg s.d.	European	m	5 NM	+/- PAR, 20 mg q.d. po
Brøsen 1993	po	100 mg s.d.	European	m	4 fast NM	+/- PAR, 20 mg q.d. po
Aldermann 1997	po	50 mg q.d.	American	m	6	+/- PAR, 20/30 mg q.d. po

Table 9: m: male, N: number of study participants, PAR: paroxetine, NM: CYP2D6 normal metabolizer, PM: CYP2D6 poor metabolizer, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

3.1.4 Paroxetine - Dextromethorphan DDGI

The paroxetine-dextromethorphan interaction was evaluated using clinical DDGI studies listed in [Table 11](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Storelli 2018	po	5 mg s.d.	European	f	16 IM	+/- PAR, 20 mg b.i.d. po
Storelli 2018	po	5 mg s.d.	European	f	17 NM	+/- PAR, 20 mg b.i.d. po
Schoedel 2012	po	30 mg b.i.d.	American	m	13	+/- PAR, 30 mg q.d. po

Table 11: b.i.d.: twice daily, f: female, IM: CYP2D6 intermediate metabolizer, m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, po: oral, pop.: population used in simulations, QUI: quinidine, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

3.2 DDGI Concentration-Time Profiles

The following section shows concentration-time profiles of the victim drugs of the simulated DD(G)I studies in comparison to observed data.

3.2.1 Paroxetine - Atomoxetine DDGI

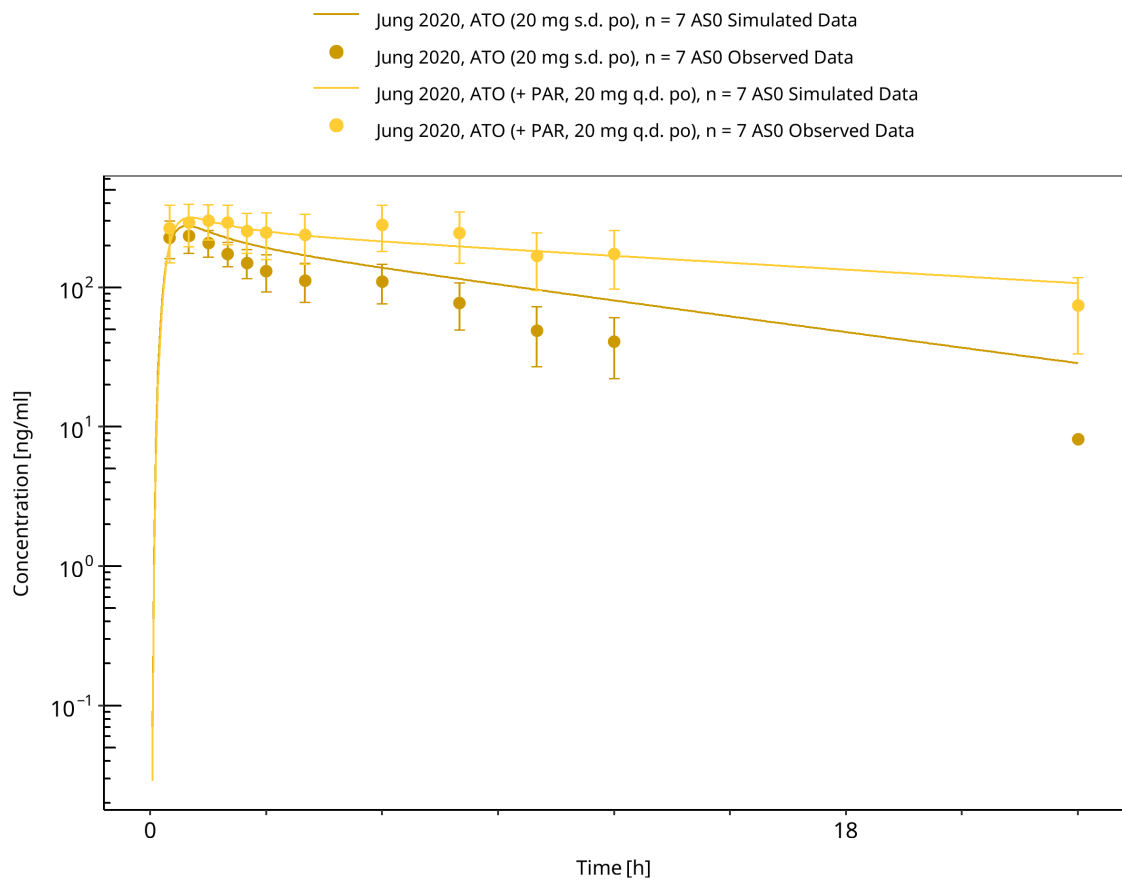


Figure 3-6: Paroxetine - Atomoxetine DDGI

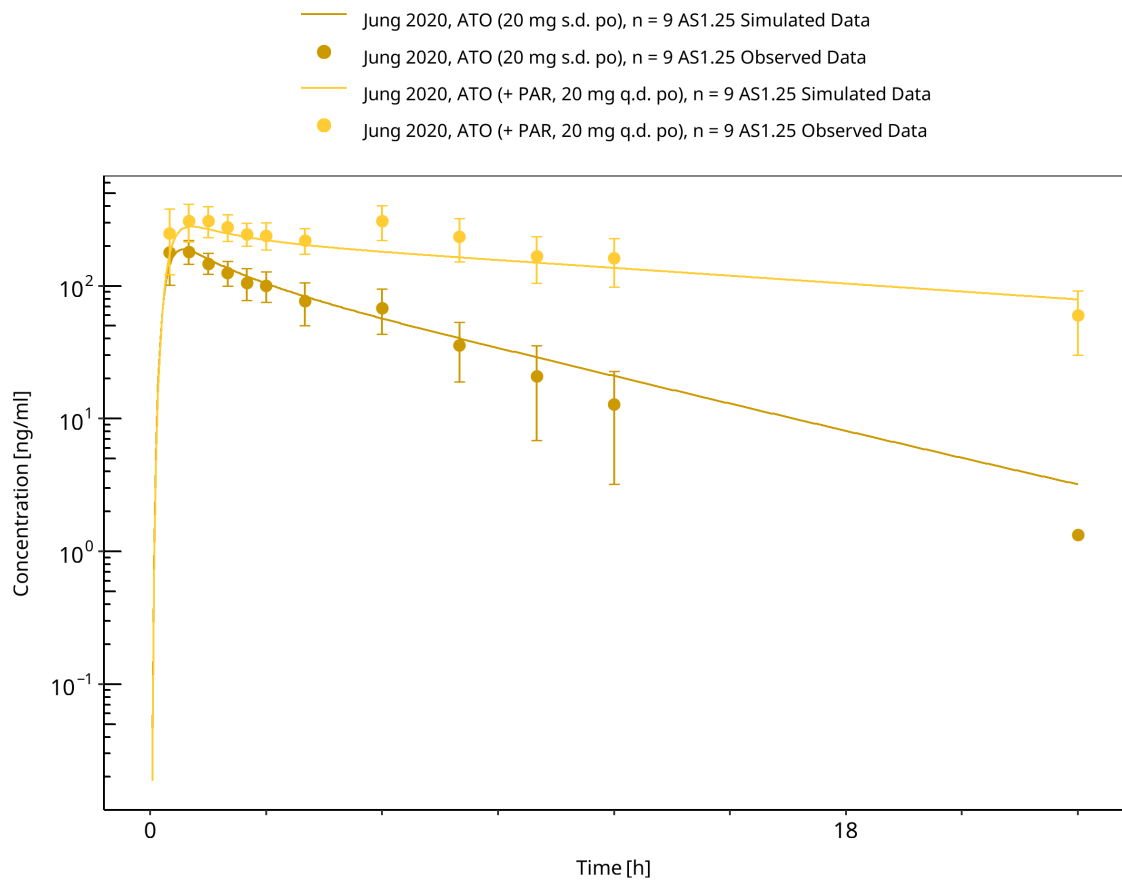


Figure 3-7: Paroxetine - Atomoxetine DDGI

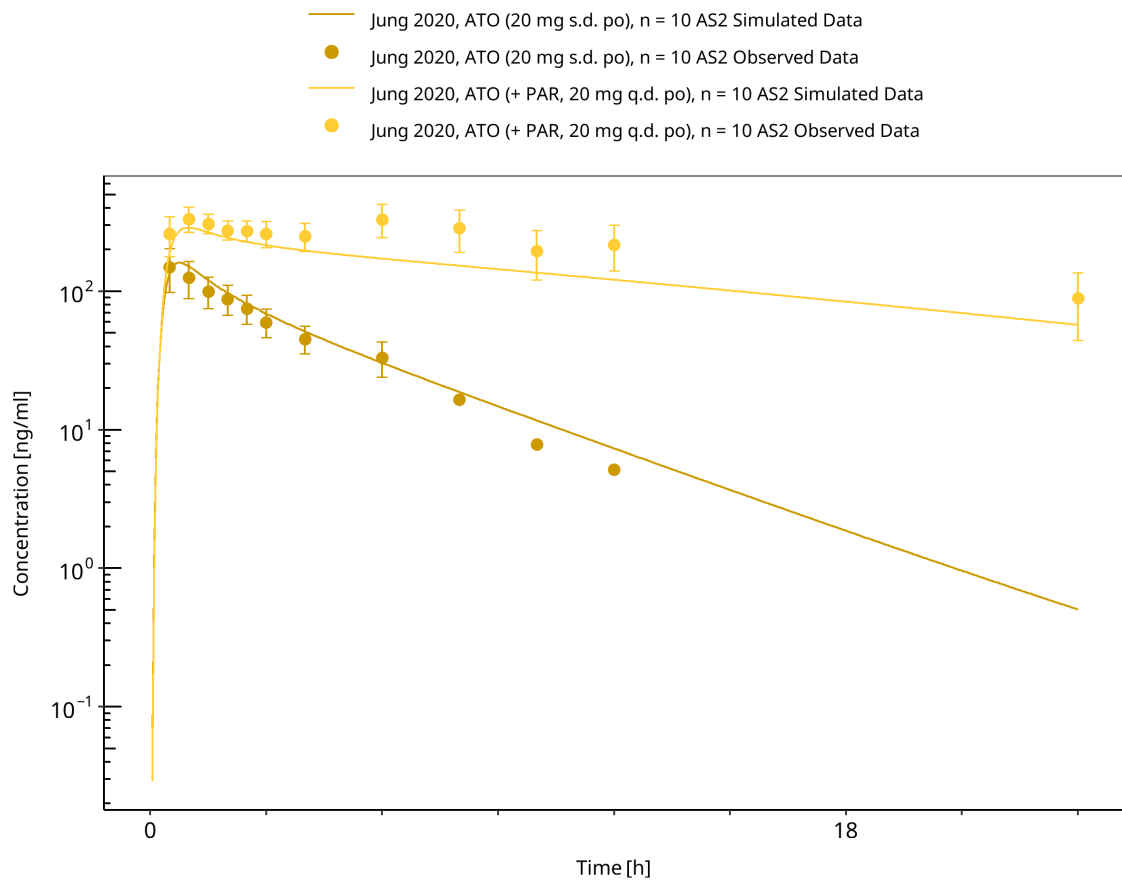


Figure 3-8: Paroxetine - Atomoxetine DDGI

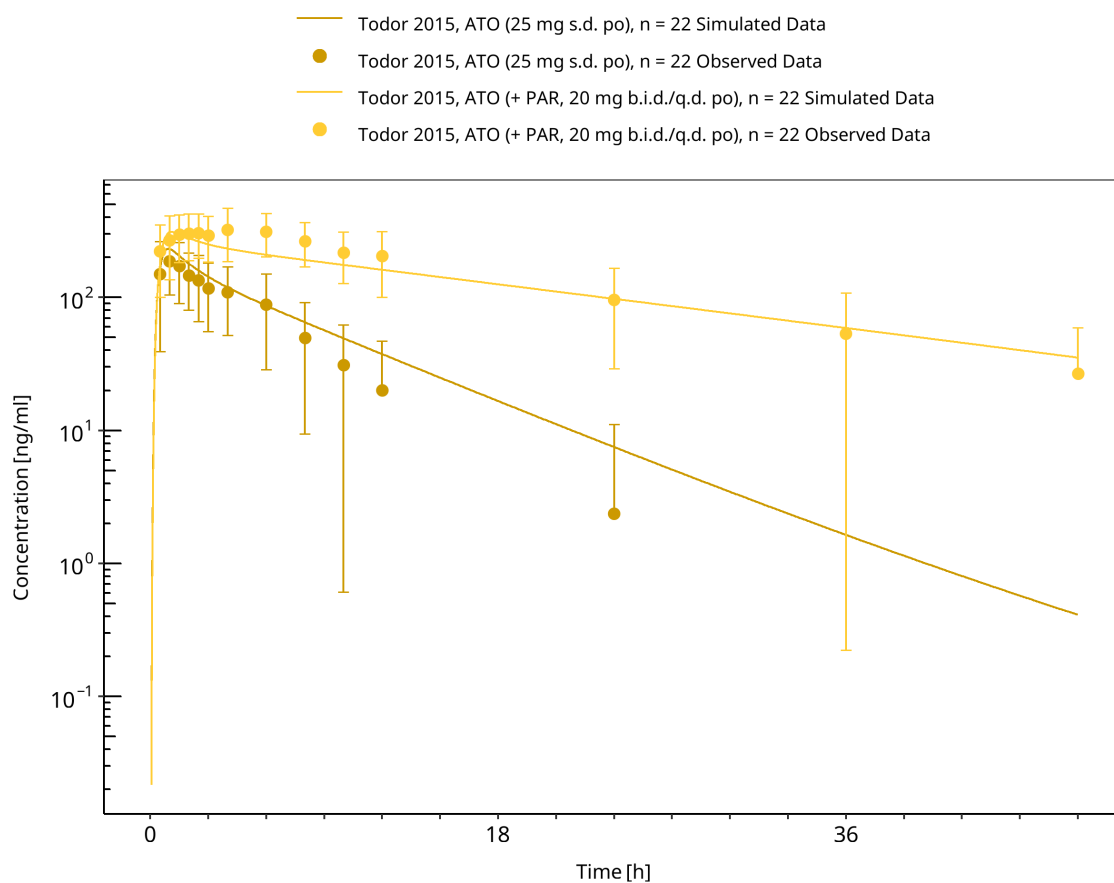


Figure 3-9: Paroxetine - Atomoxetine DDI

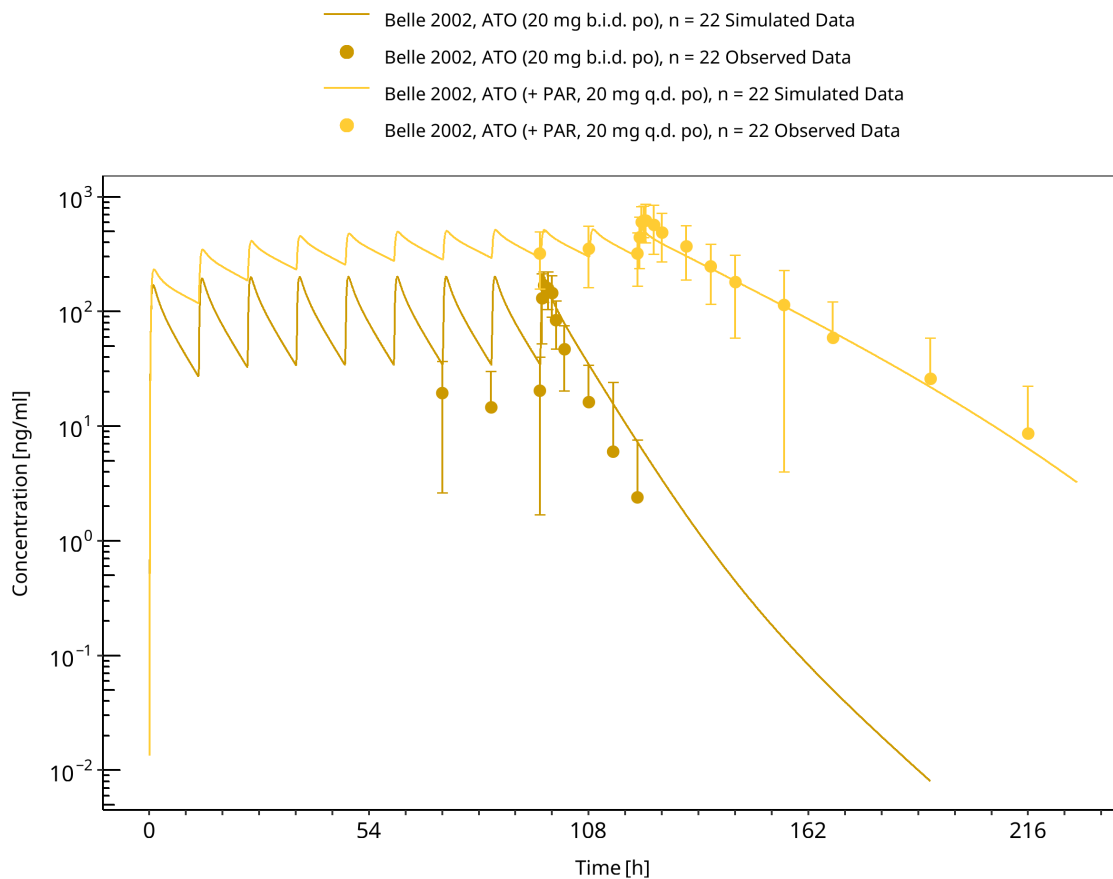


Figure 3-10: Paroxetine - Atomoxetine DDI

3.2.2 Paroxetine - Clomiphene DDGI

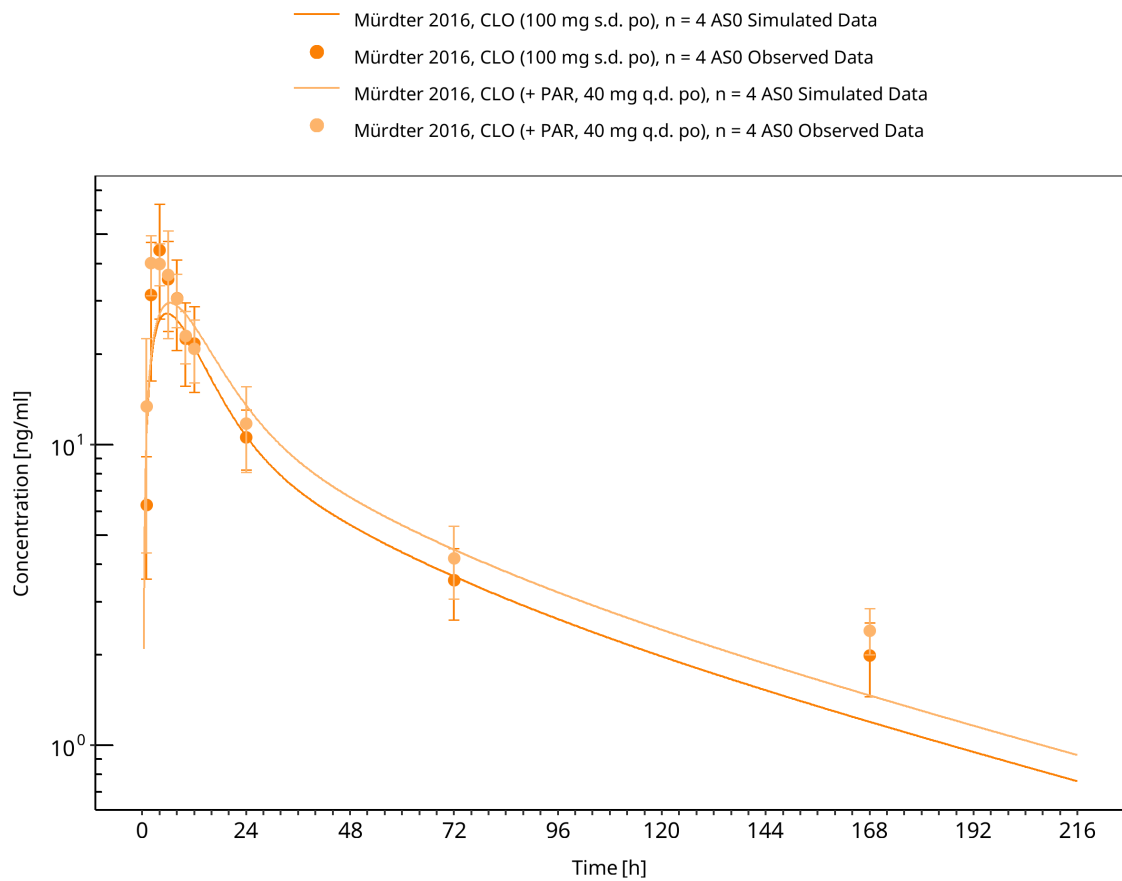


Figure 3-11: Paroxetine - Clomiphene DDGI

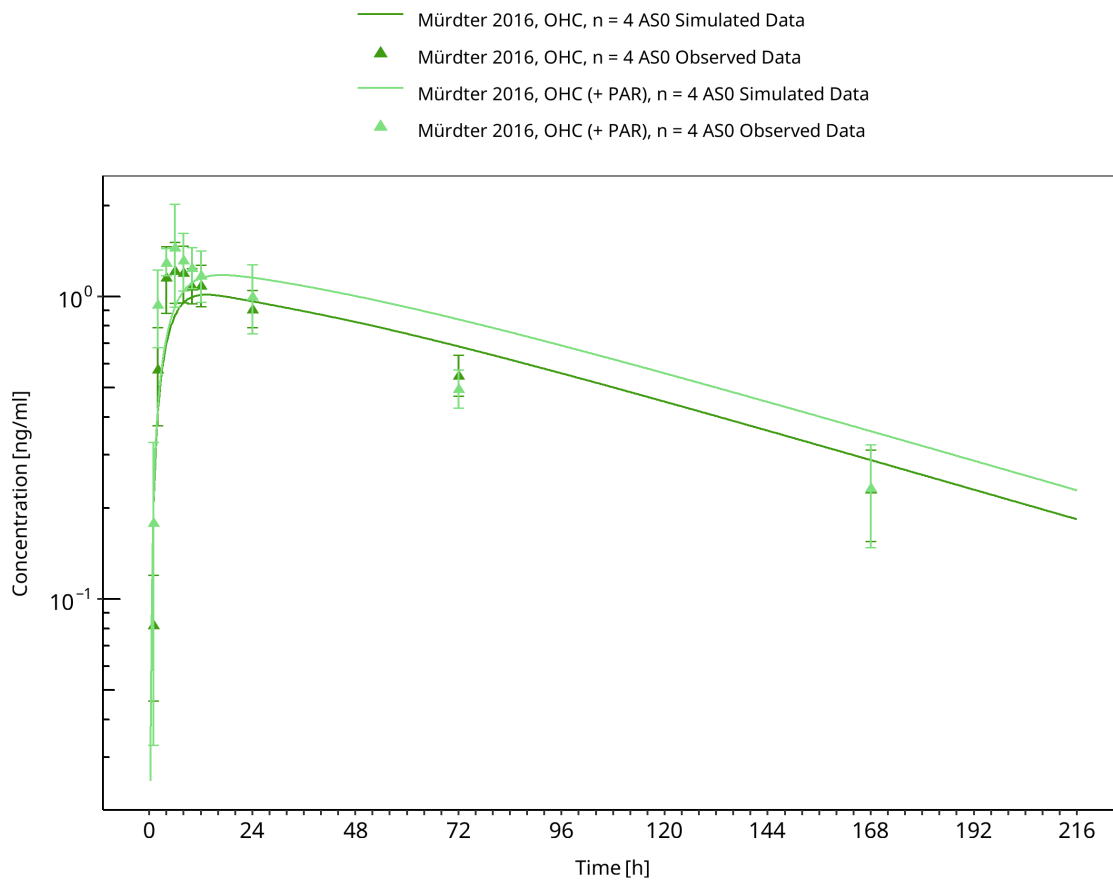


Figure 3-12: Paroxetine - Clomiphen DDGI

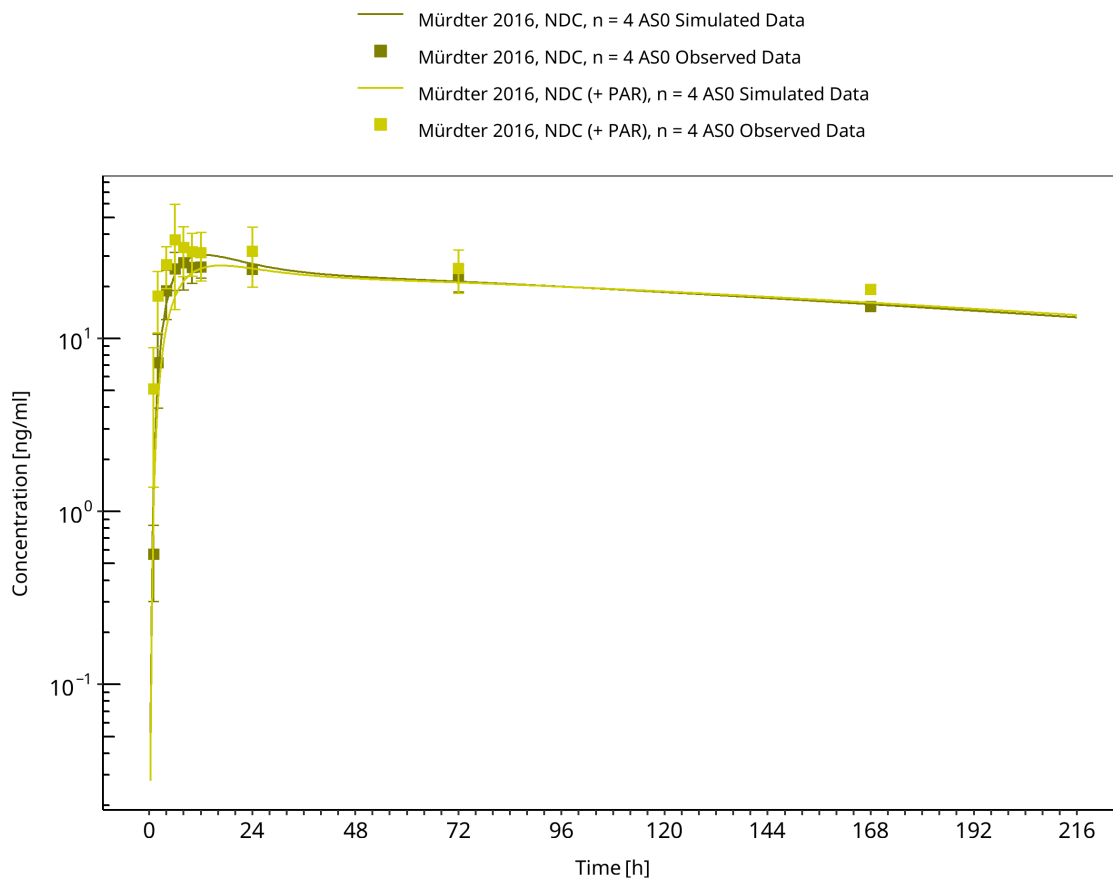


Figure 3-13: Paroxetine - Clomiphene DDGI

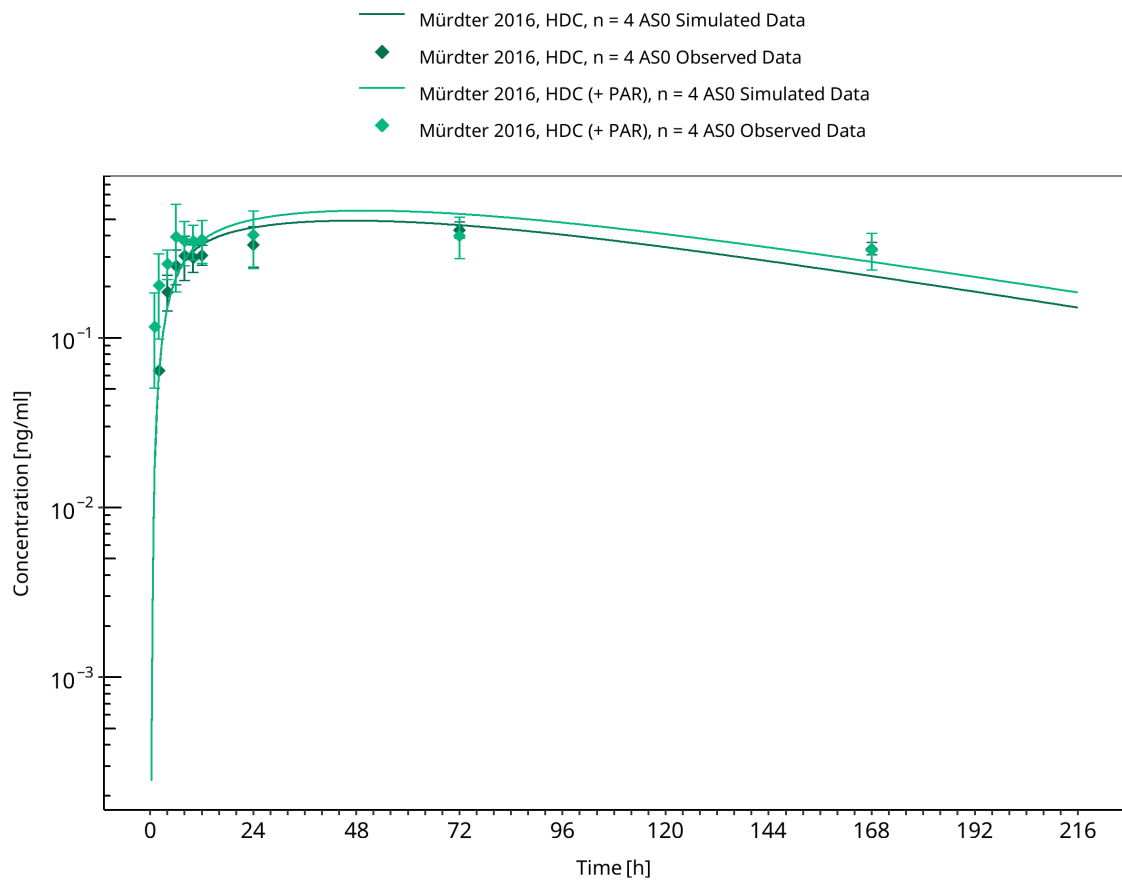


Figure 3-14: Paroxetine - Clomiphene DDGI

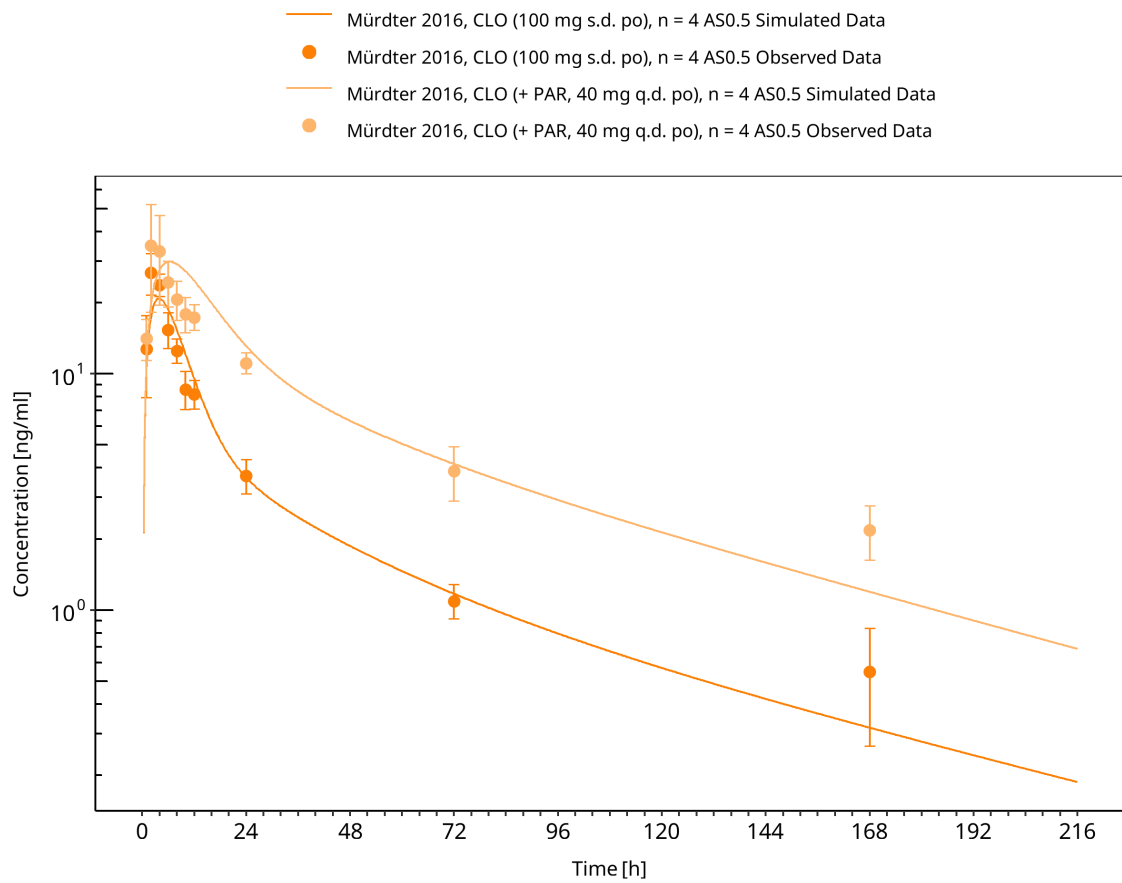


Figure 3-15: Paroxetine - Clomiphene DDGI

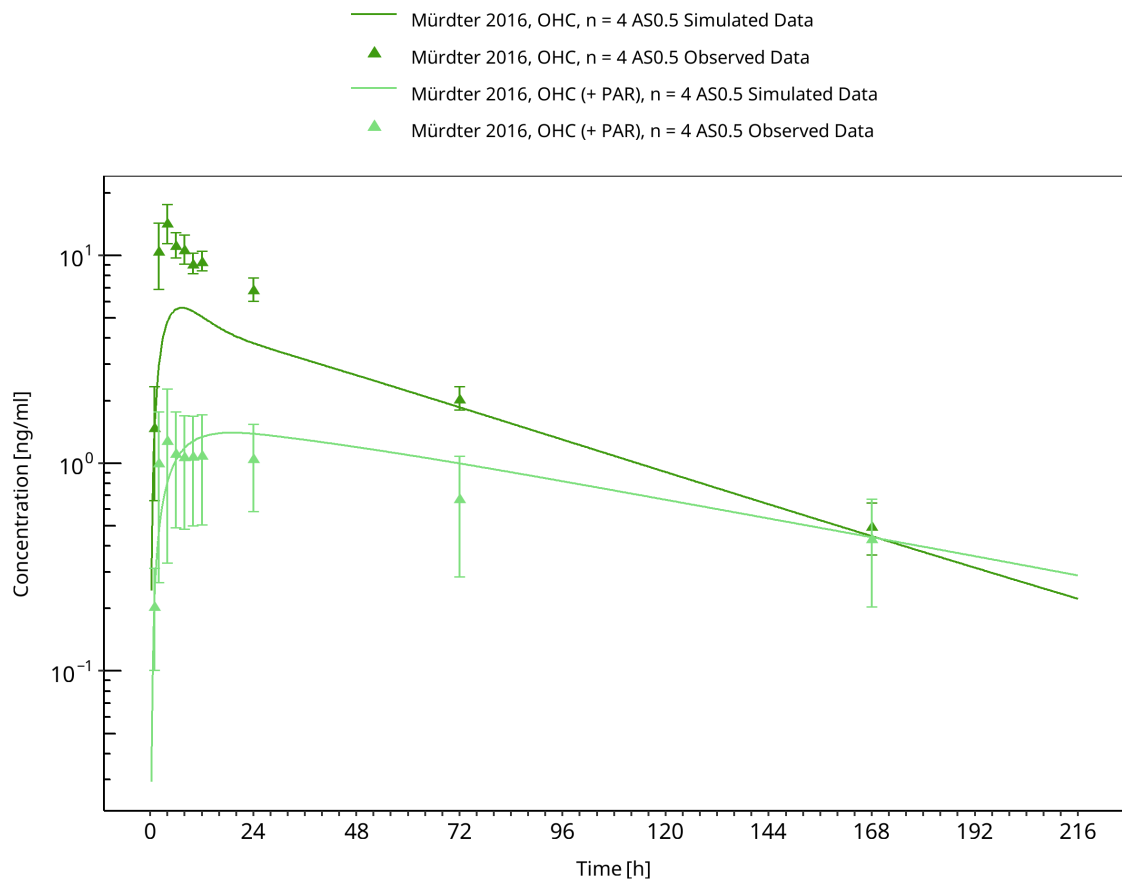


Figure 3-16: Paroxetine - Clomiphene DDGI

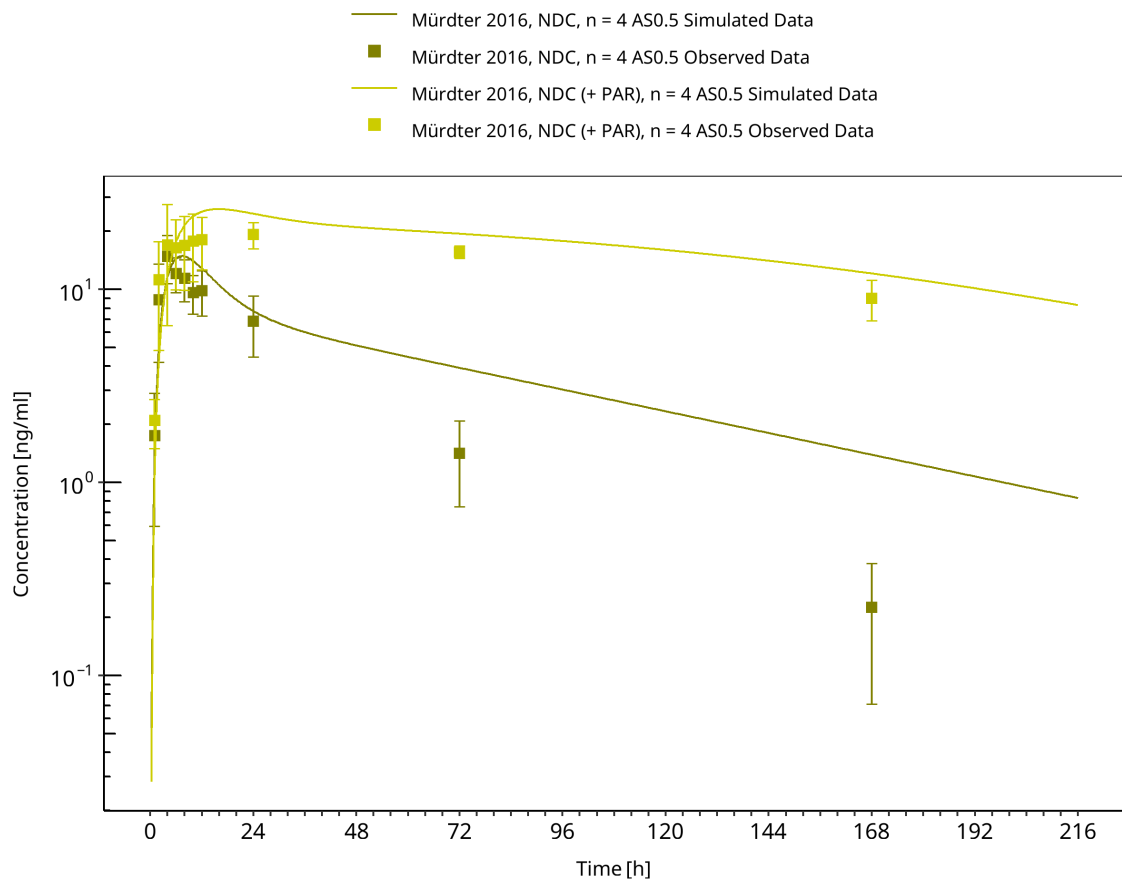


Figure 3-17: Paroxetine - Clomiphene DDGI

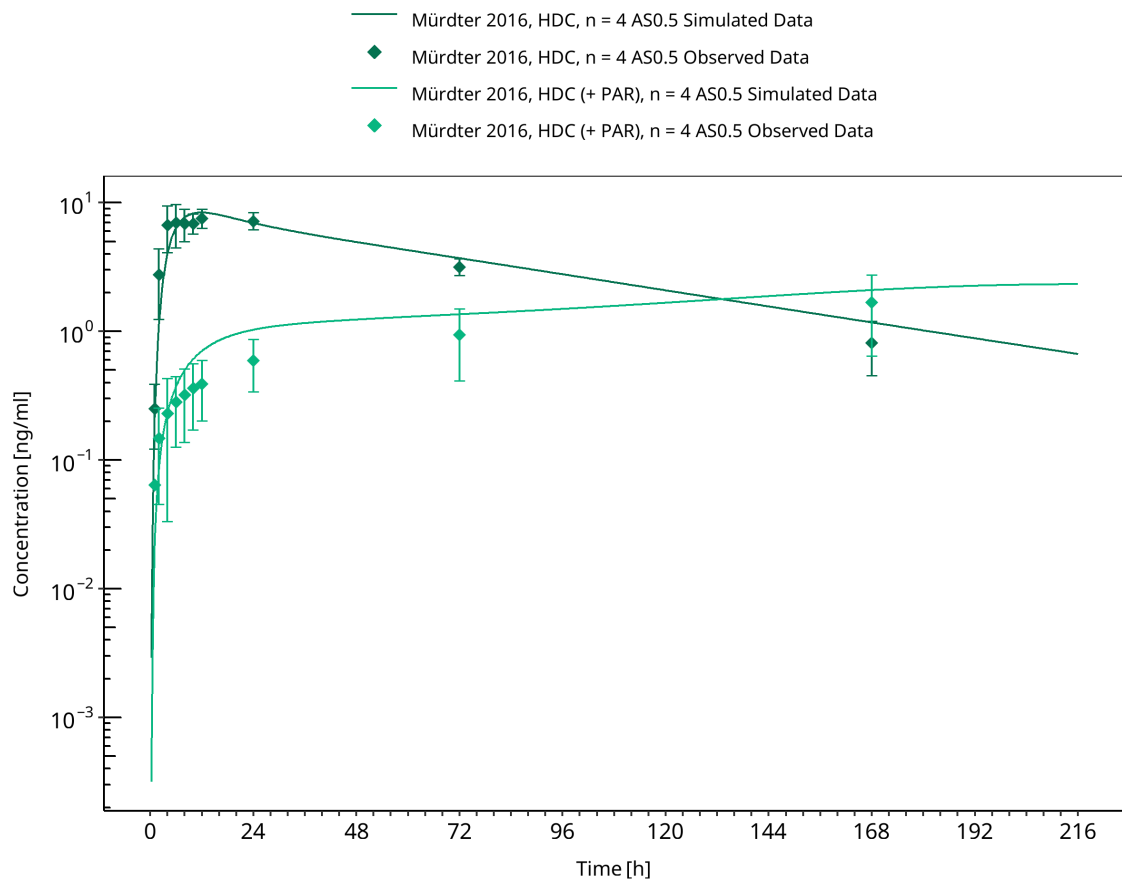


Figure 3-18: Paroxetine - Clomiphenes DDGI

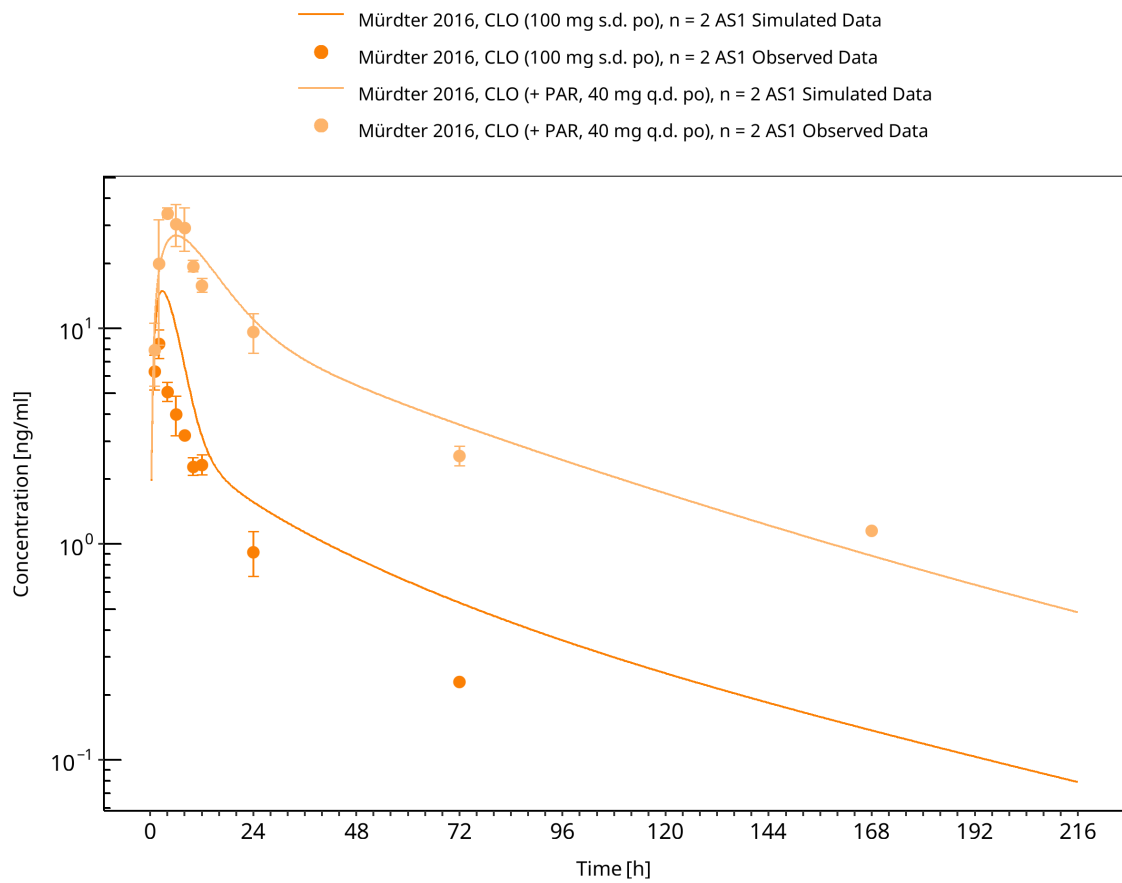


Figure 3-19: Paroxetine - Clomiphene DDGI

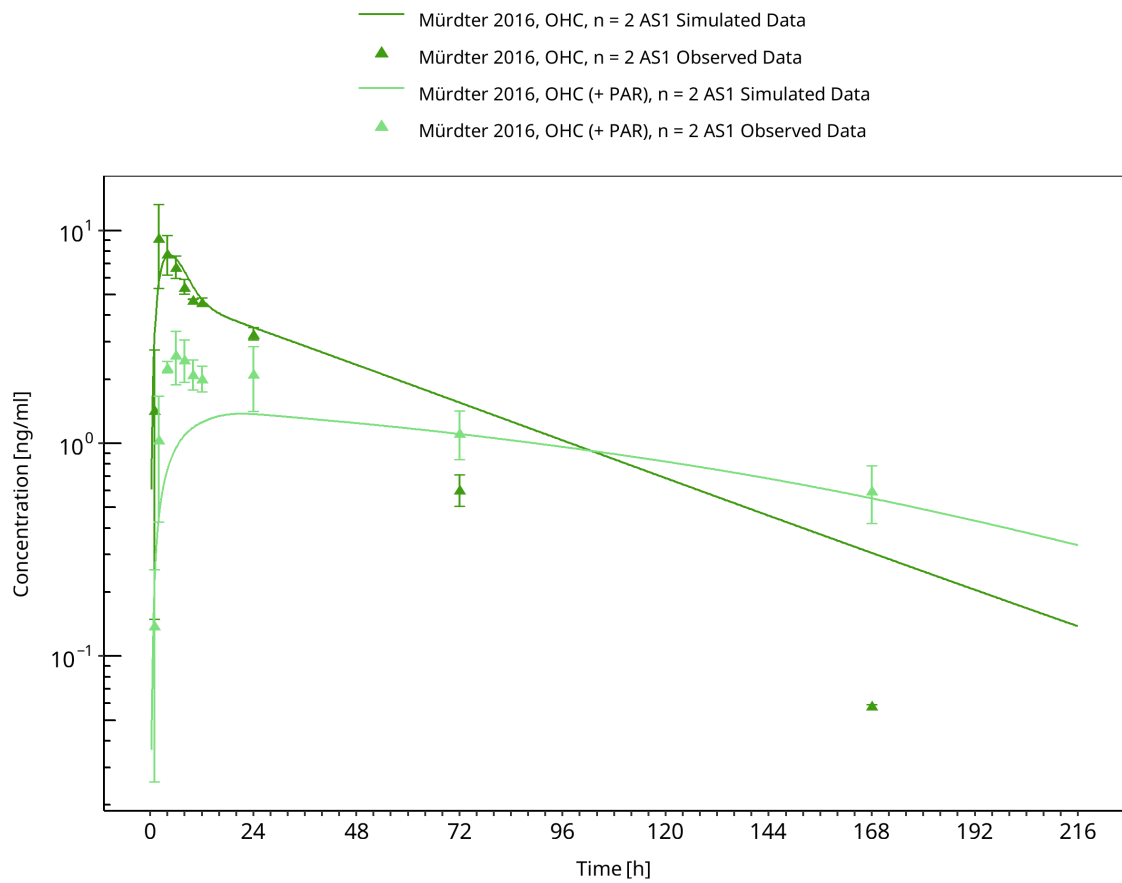


Figure 3-20: Paroxetine - Clomiphen DDGI

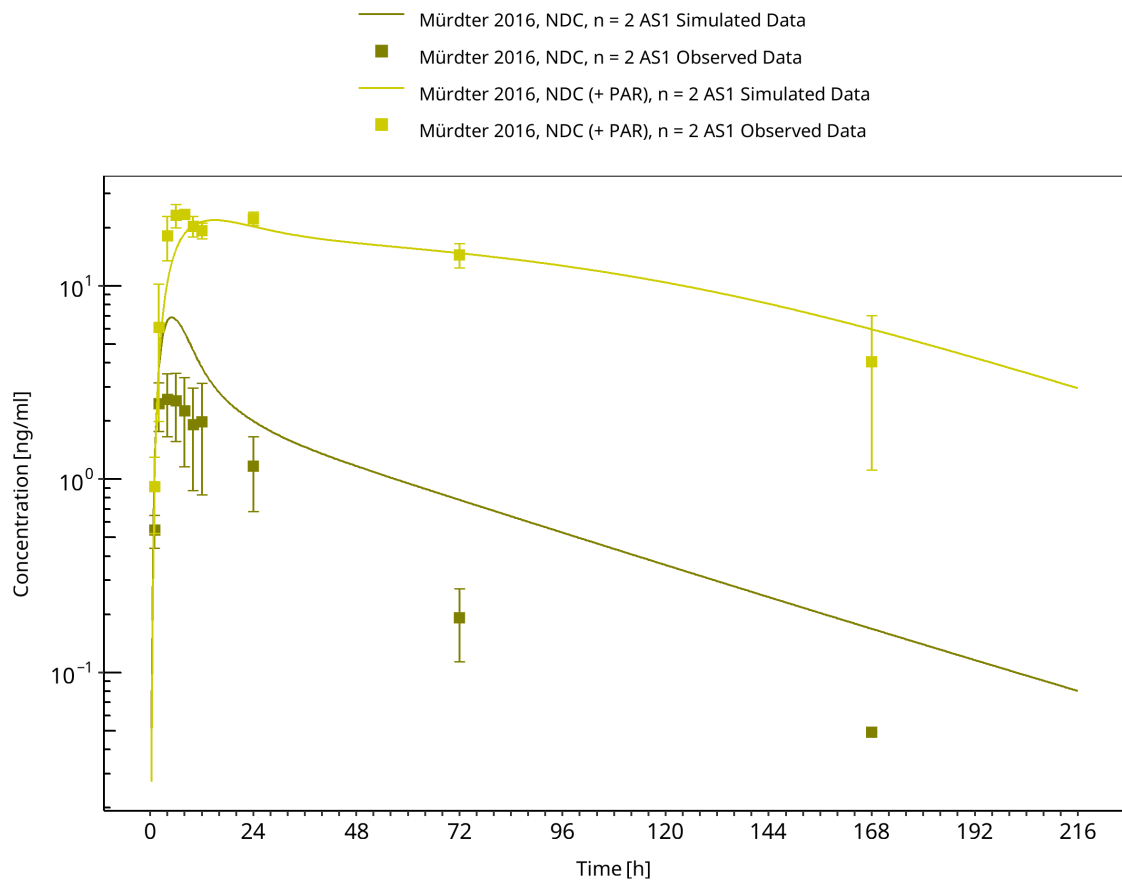


Figure 3-21: Paroxetine - Clomiphene DDGI

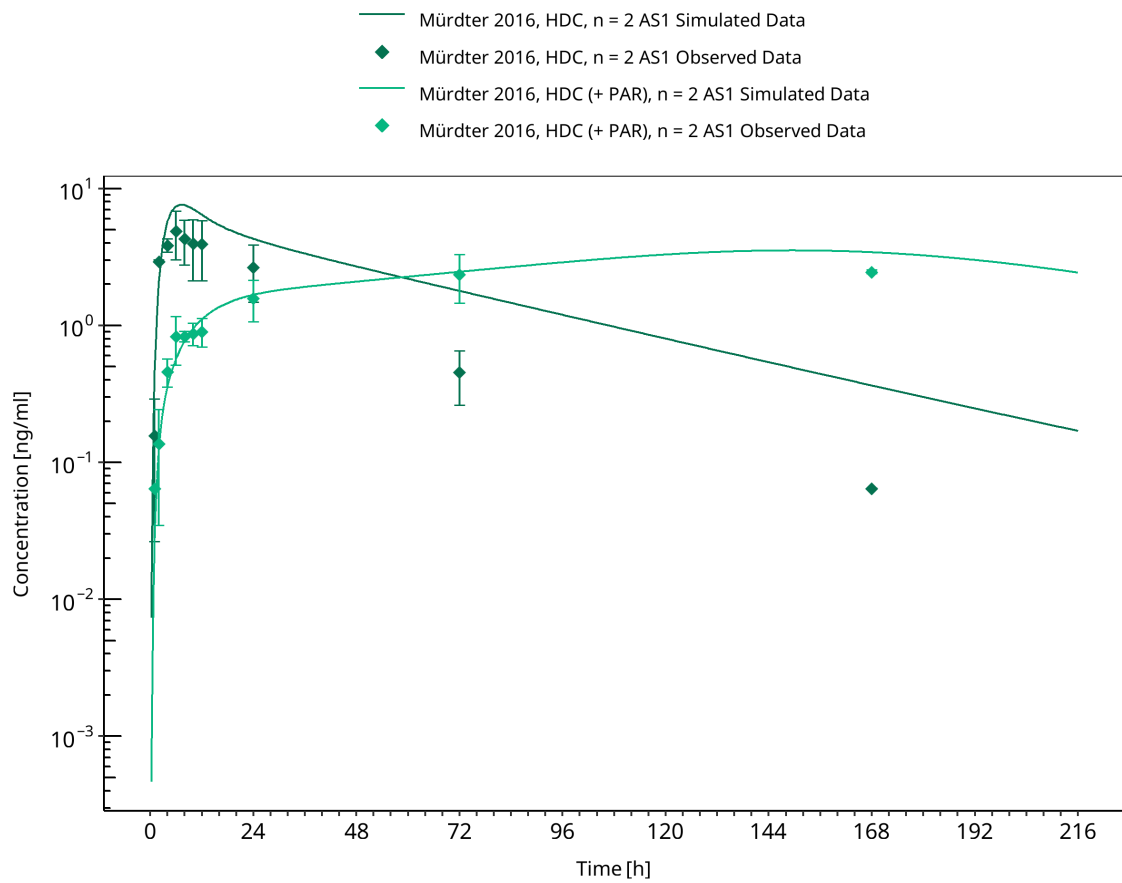


Figure 3-22: Paroxetine - Clomiphene DDGI

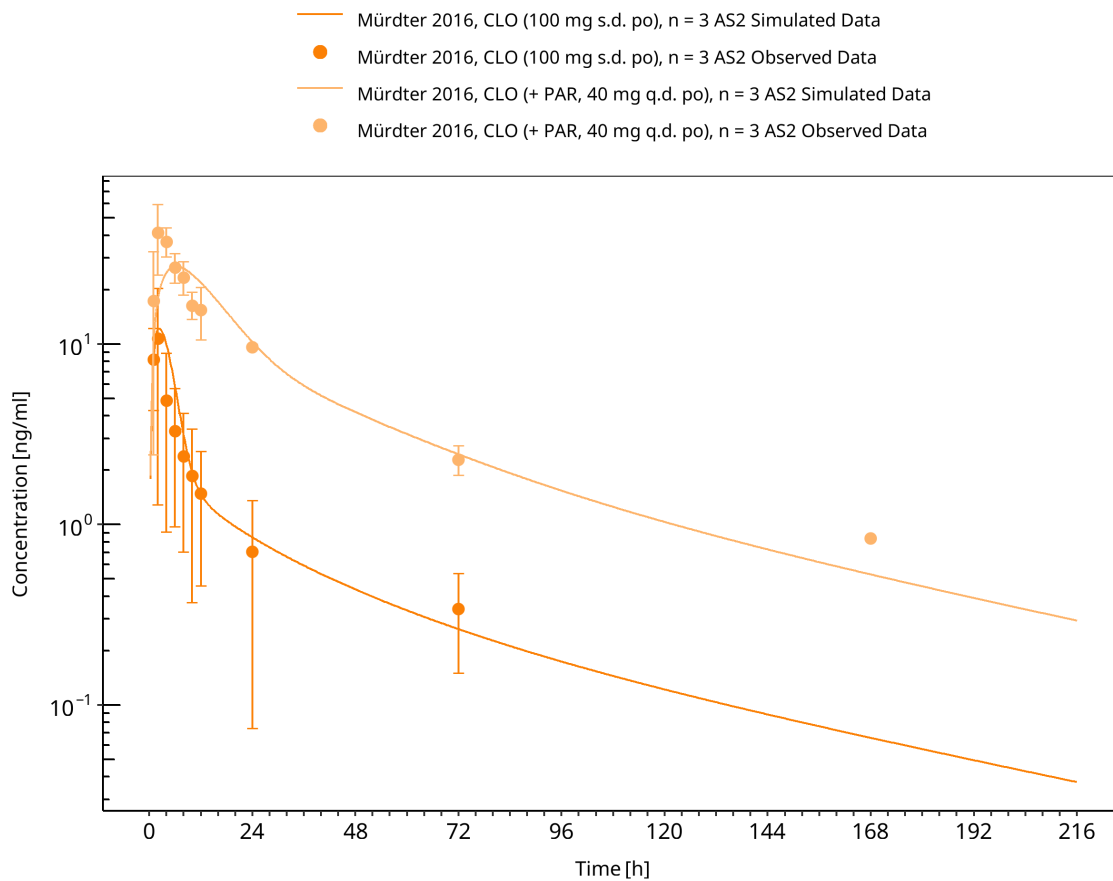


Figure 3-23: Paroxetine - Clomiphene DDGI

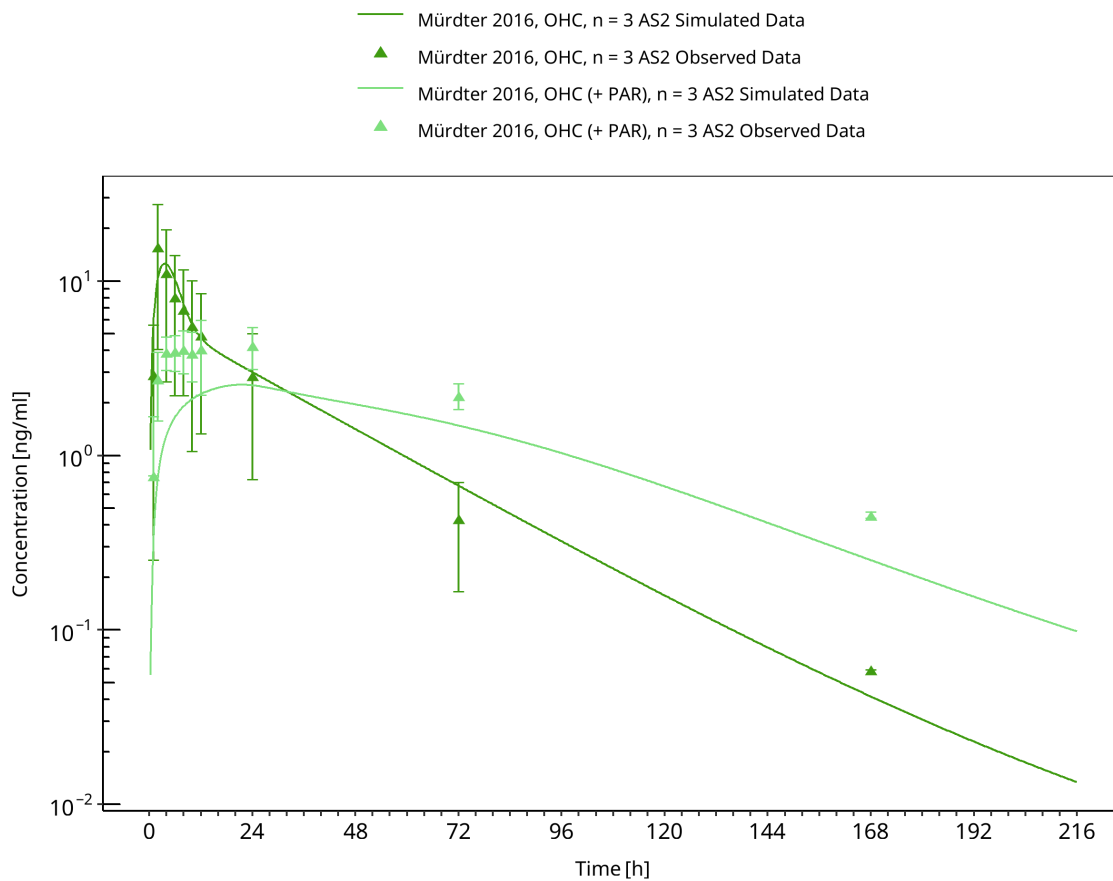


Figure 3-24: Paroxetine - Clomiphene DDGI

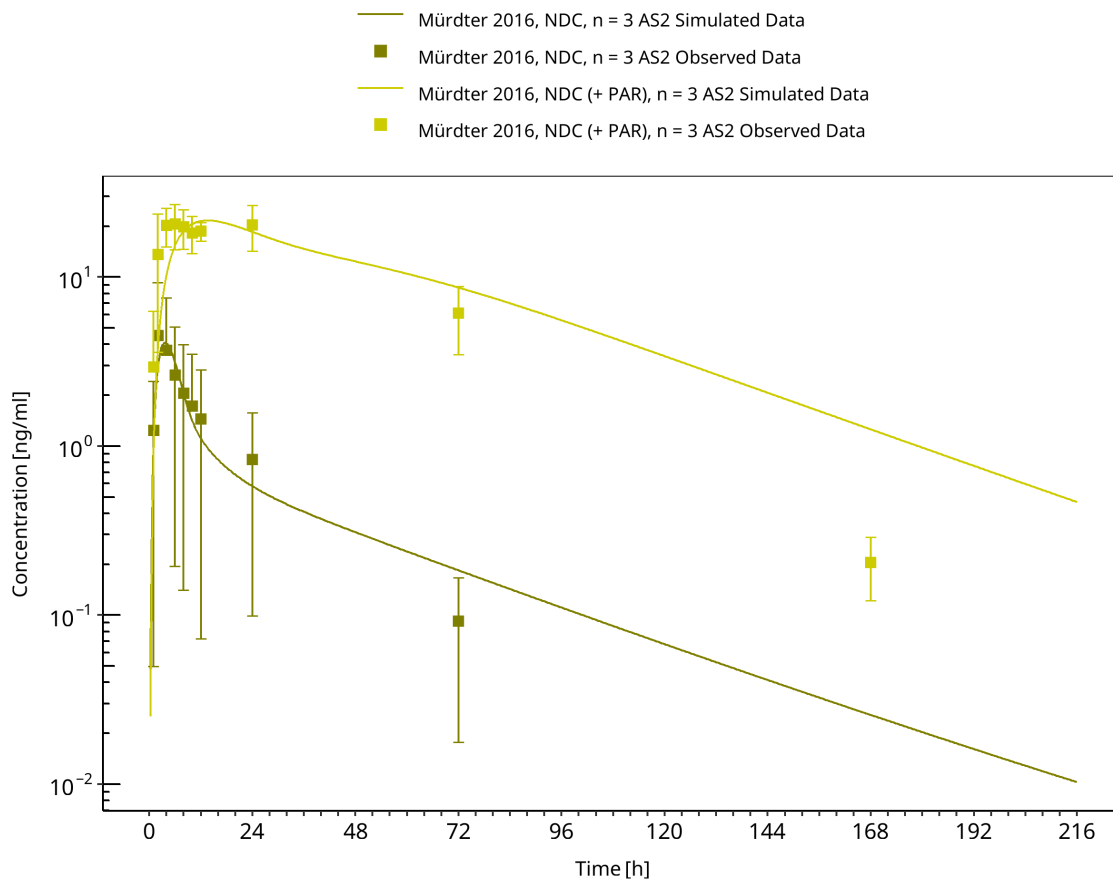


Figure 3-25: Paroxetine - Clomiphene DDGI

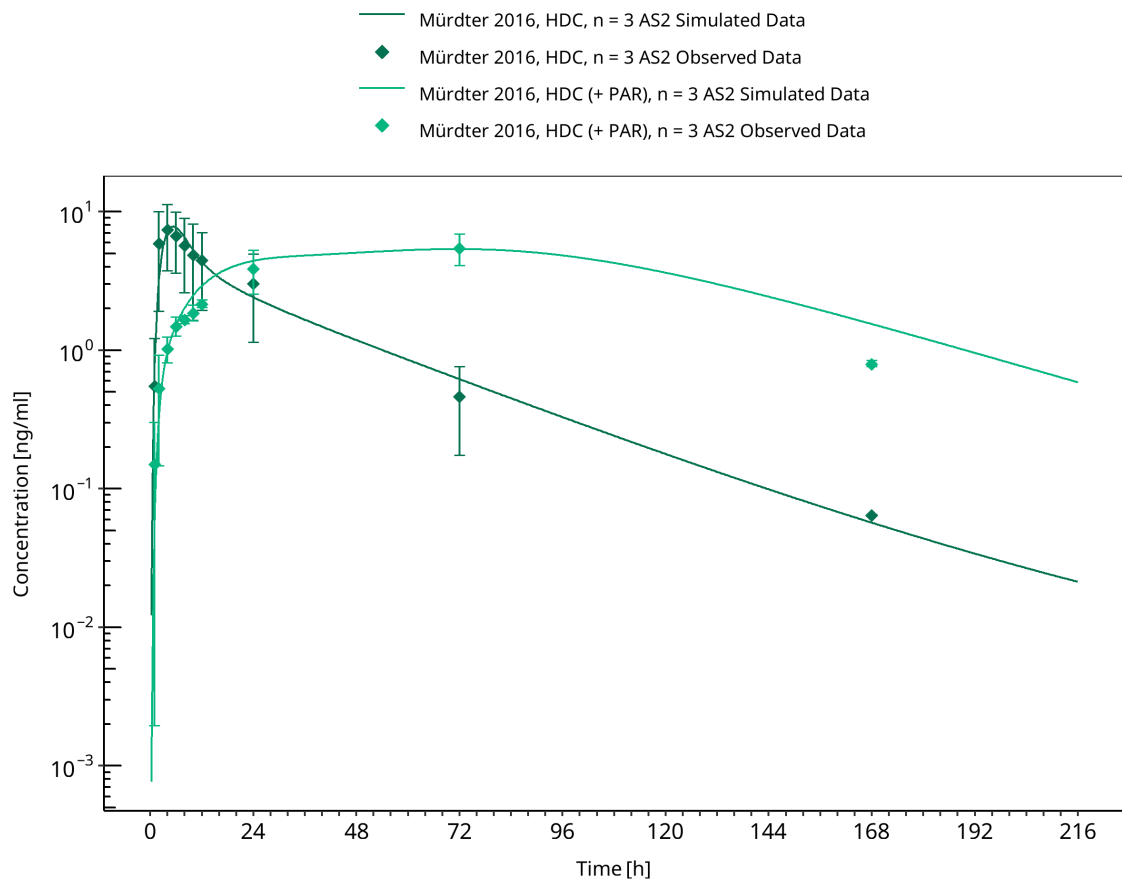


Figure 3-26: Paroxetine - Clomiphene DDGI

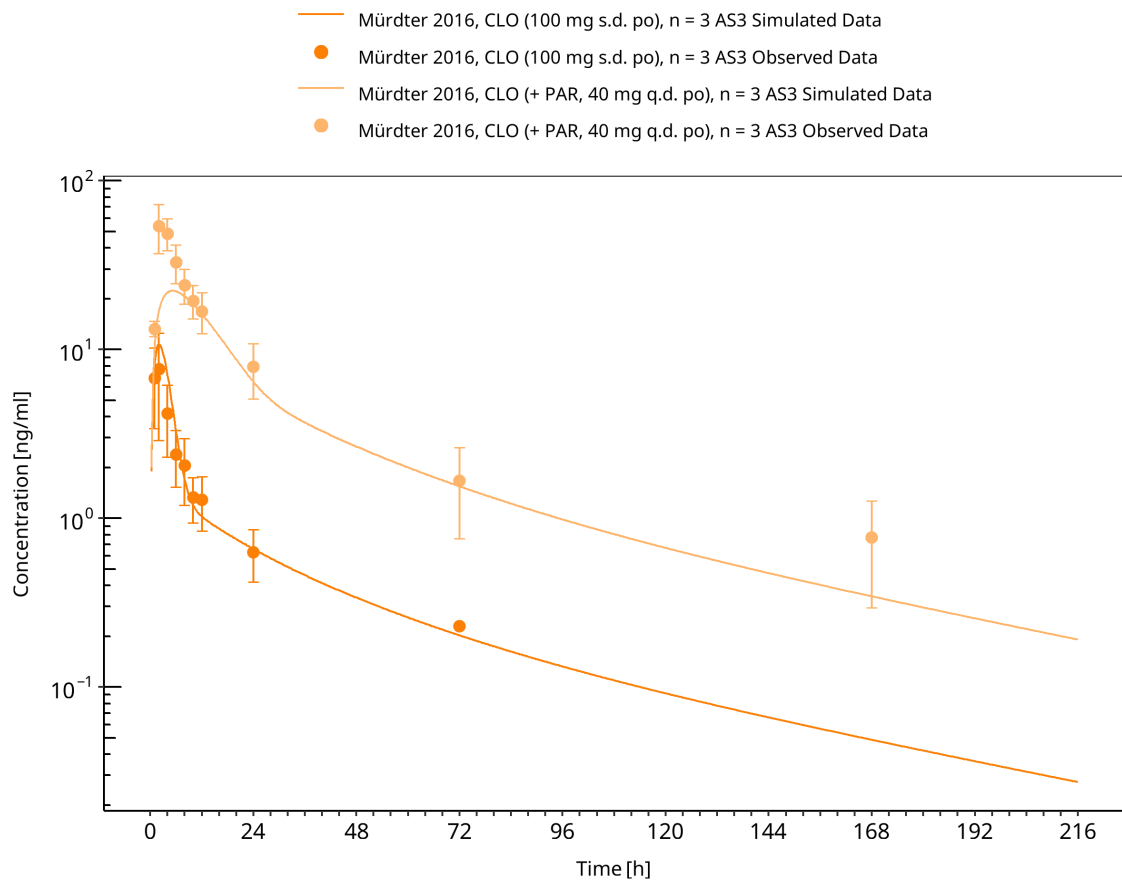


Figure 3-27: Paroxetine - Clomiphene DDGI

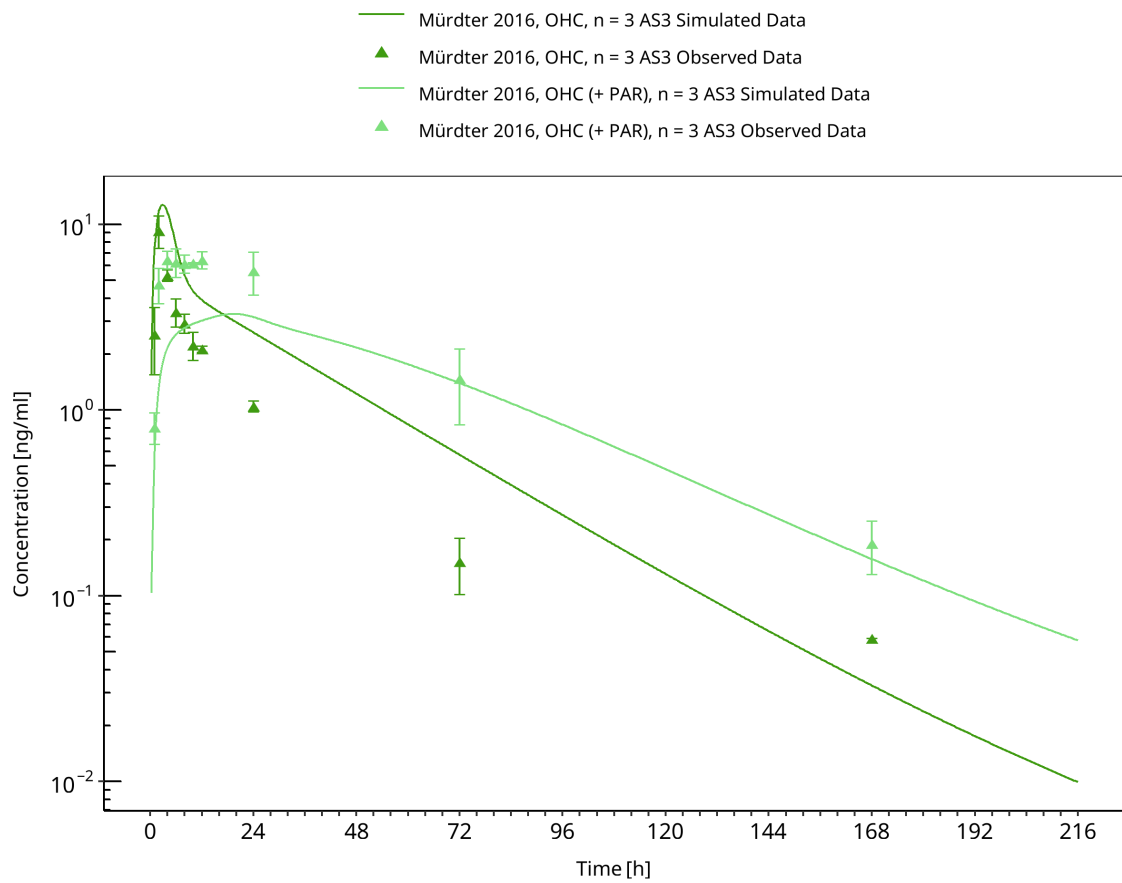


Figure 3-28: Paroxetine - Clomiphene DDGI

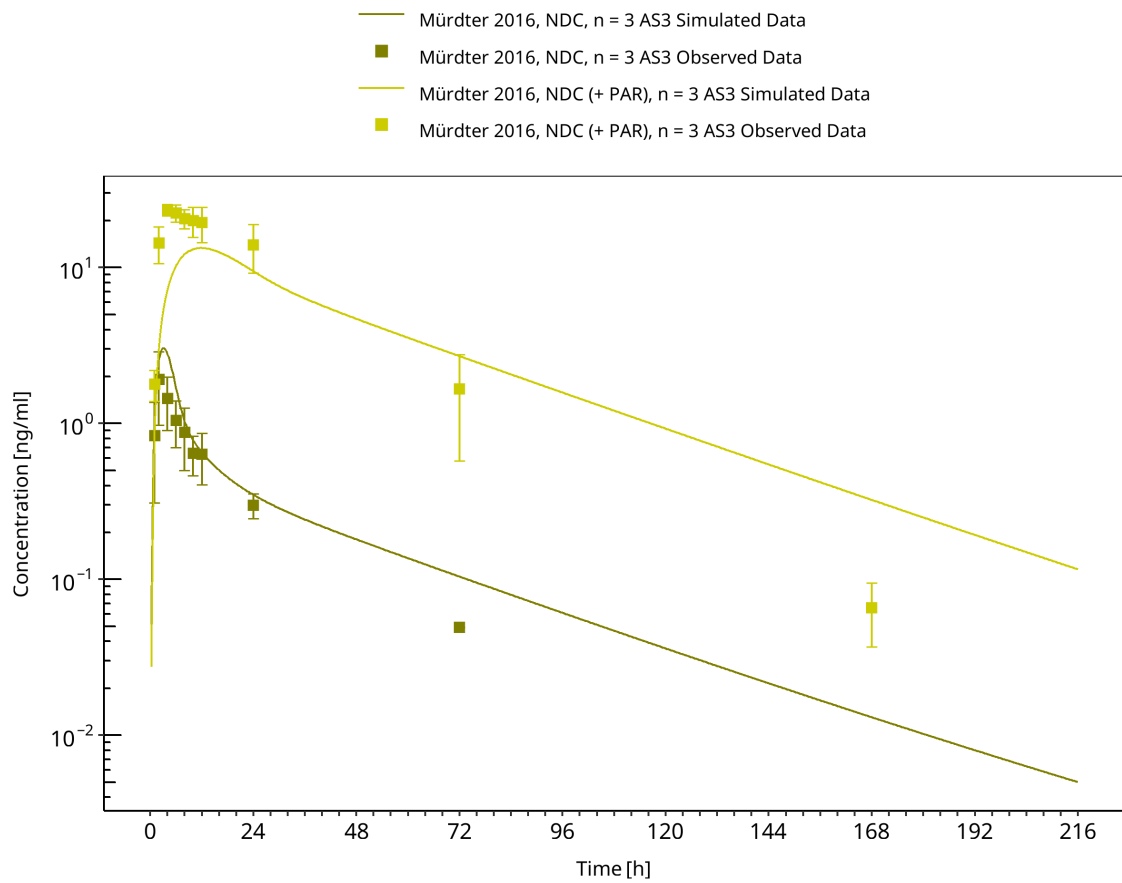


Figure 3-29: Paroxetine - Clomiphene DDGI

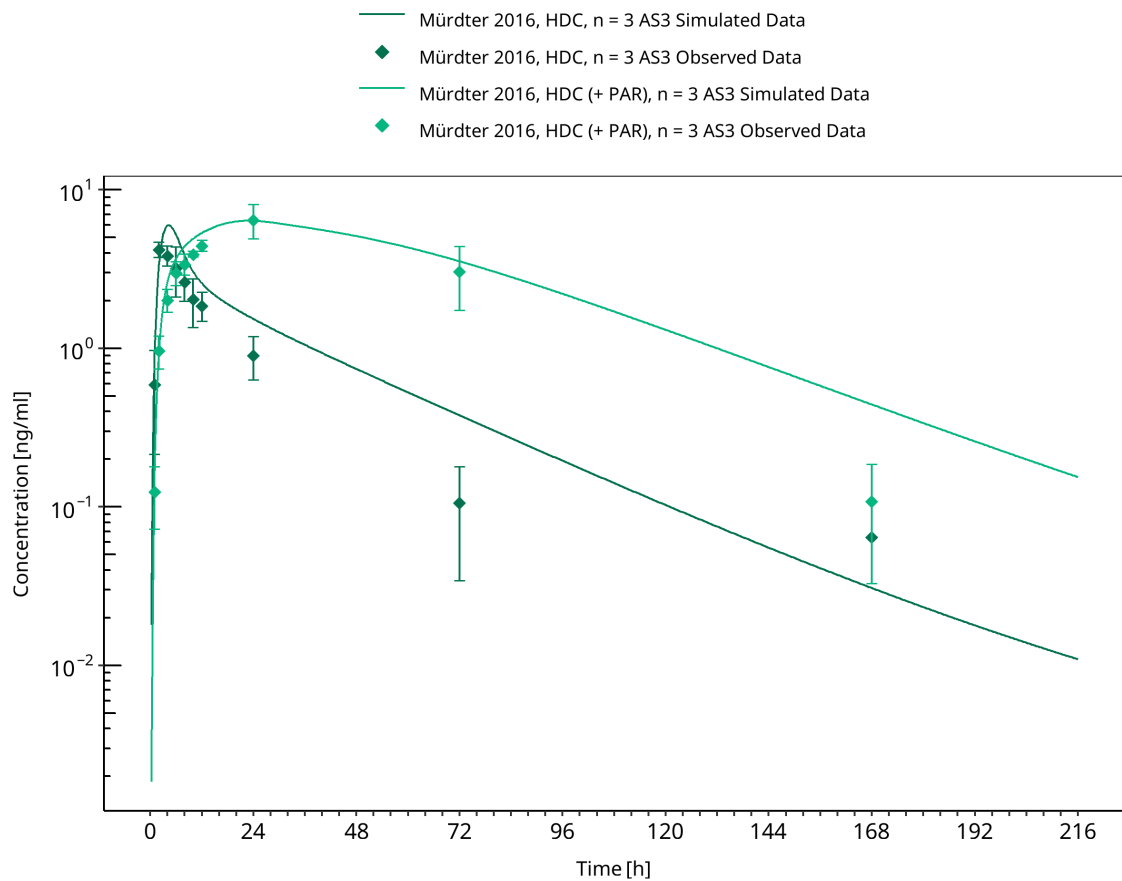


Figure 3-30: Paroxetine - Clomiphene DDGI

3.2.3 Paroxetine - Desipramine DDGI

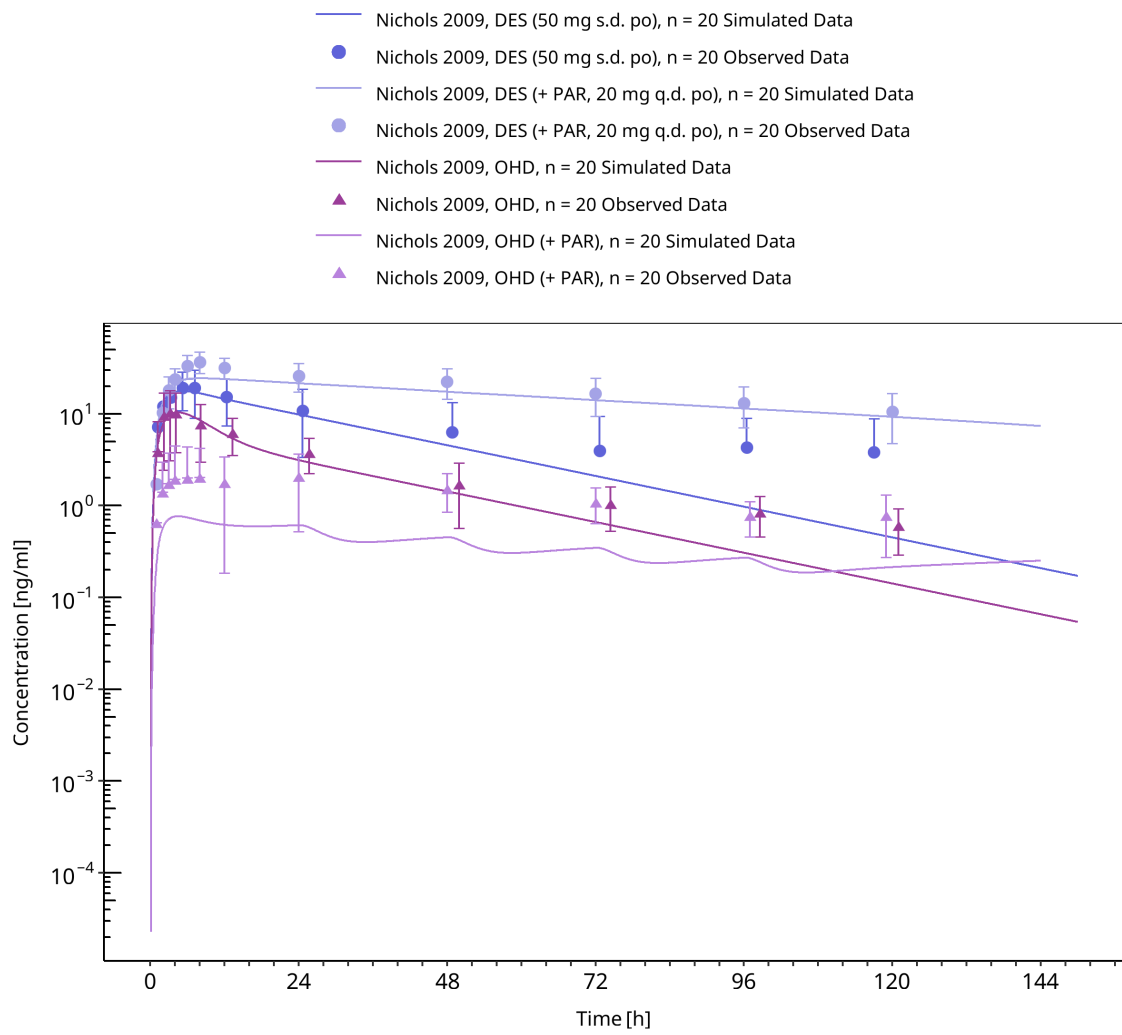


Figure 3-31: Paroxetine - Desipramine DDI

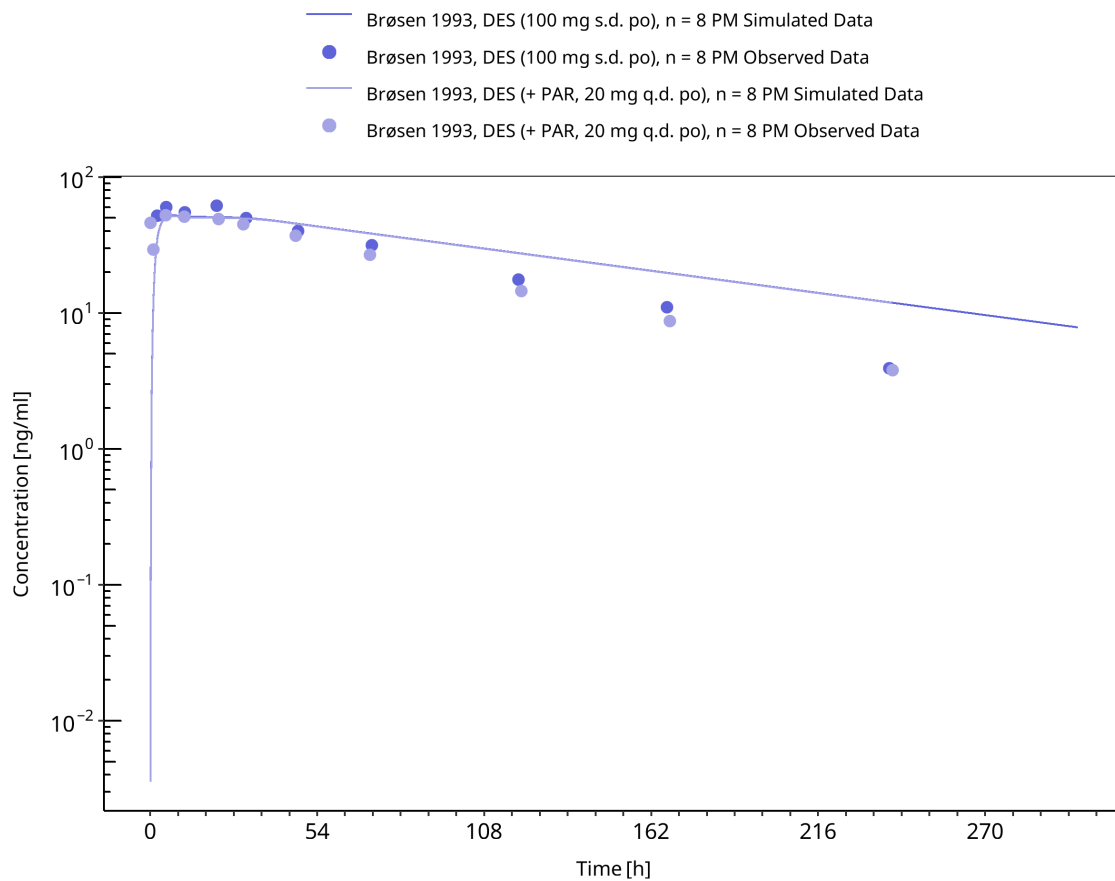


Figure 3-32: Paroxetine - Desipramine DDGI

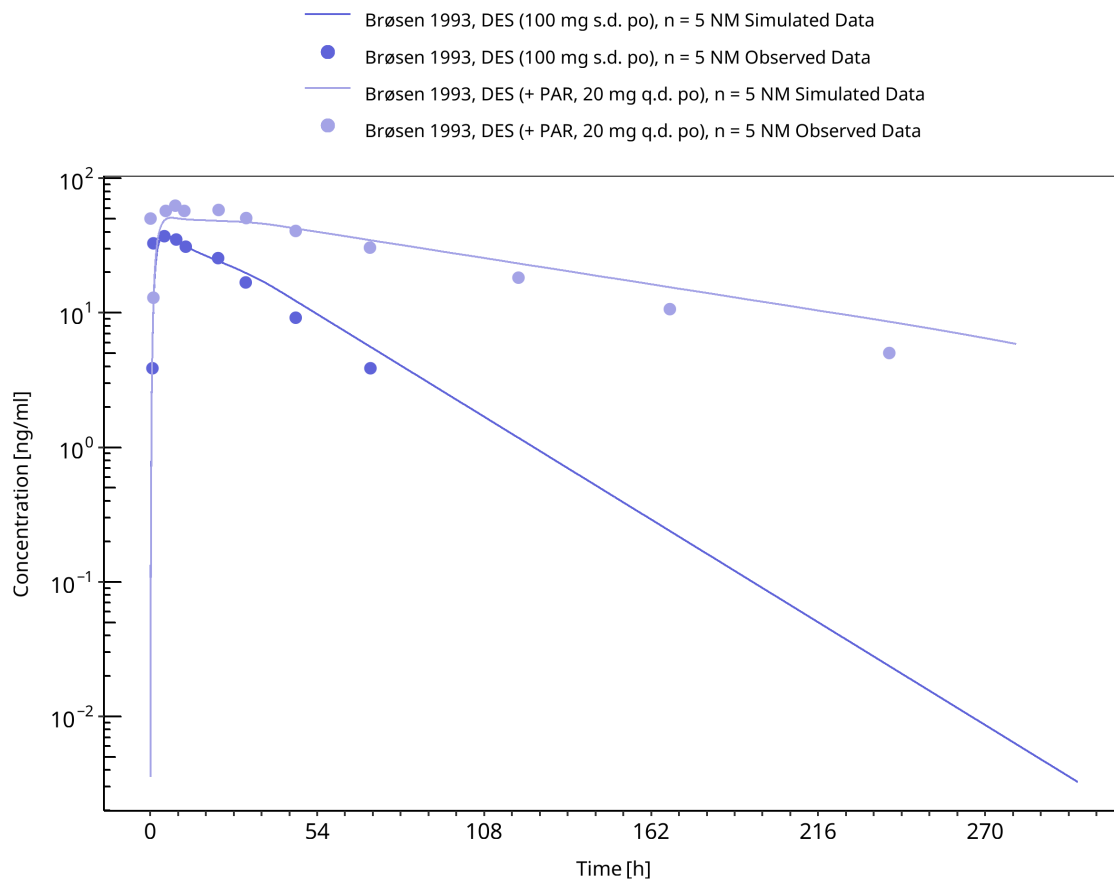


Figure 3-33: Paroxetine - Desipramine DDGI

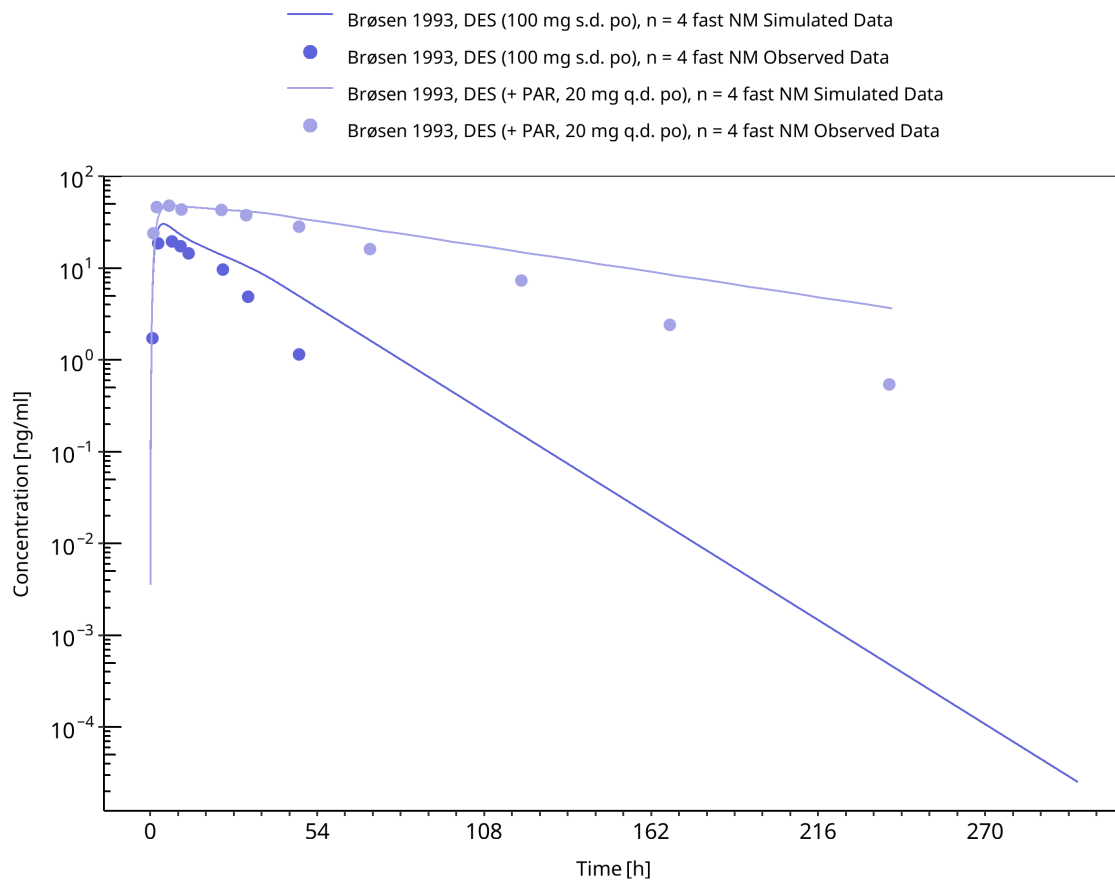


Figure 3-34: Paroxetine - Desipramine DDGI

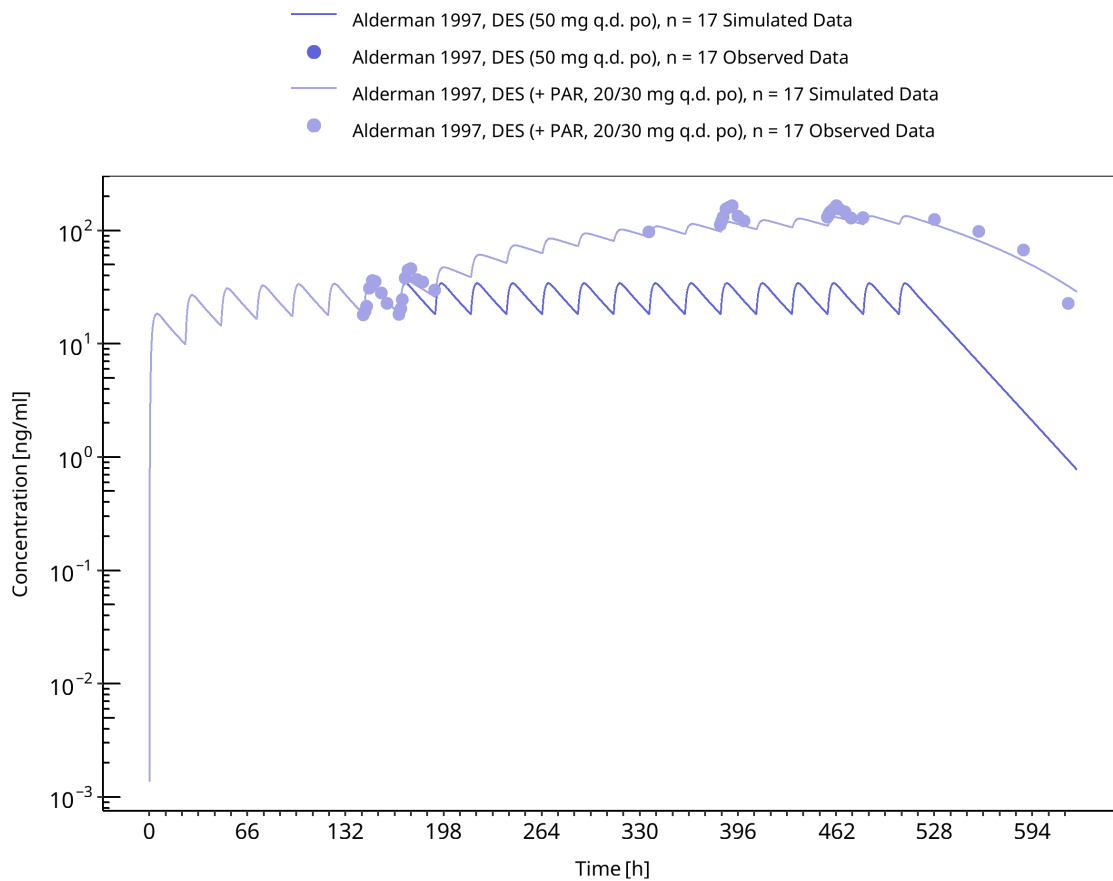


Figure 3-35: Paroxetine - Desipramine DDI

3.2.4 Paroxetine - Dextromethorphan DDGI

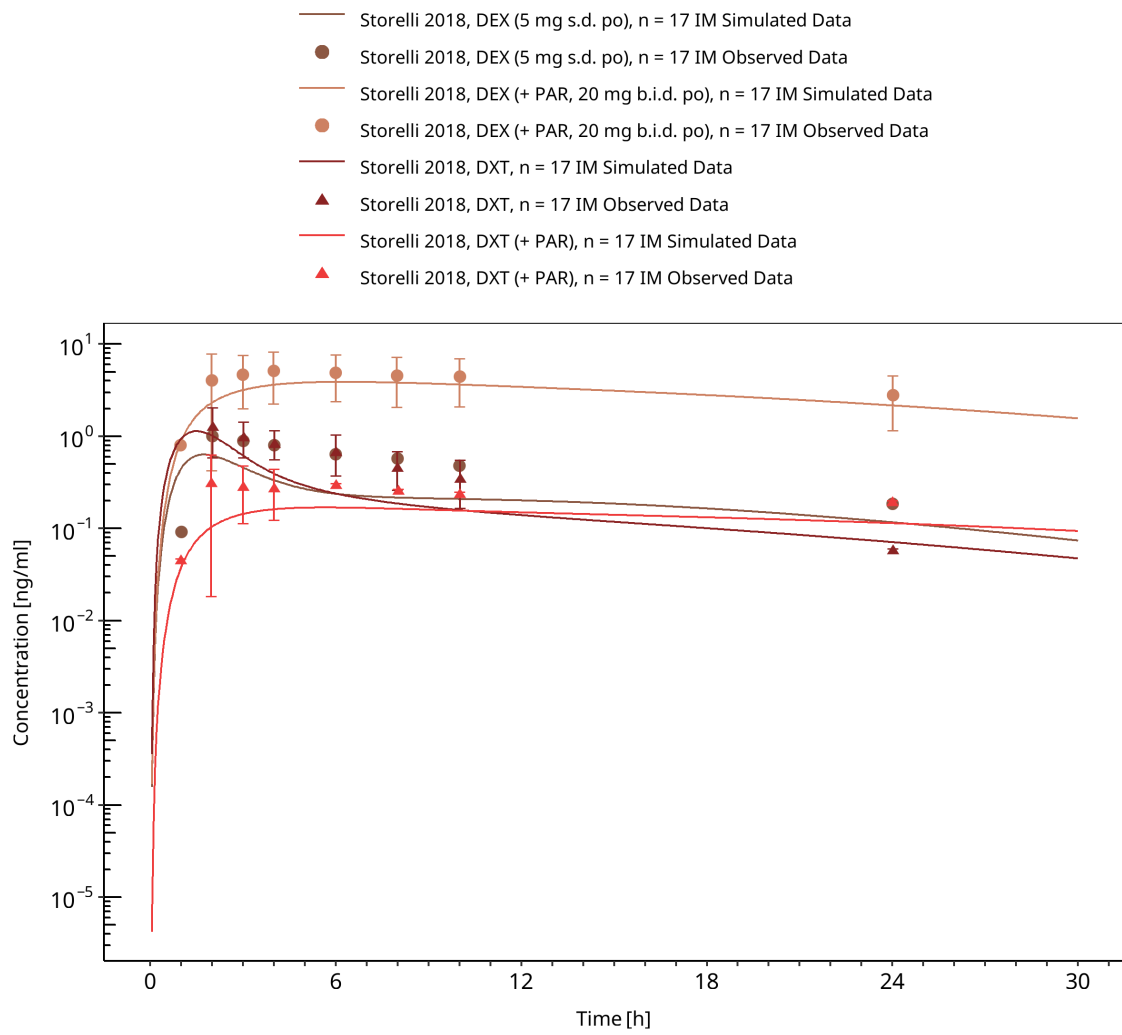


Figure 3-36: Paroxetine - Dextromethorphan DDGI

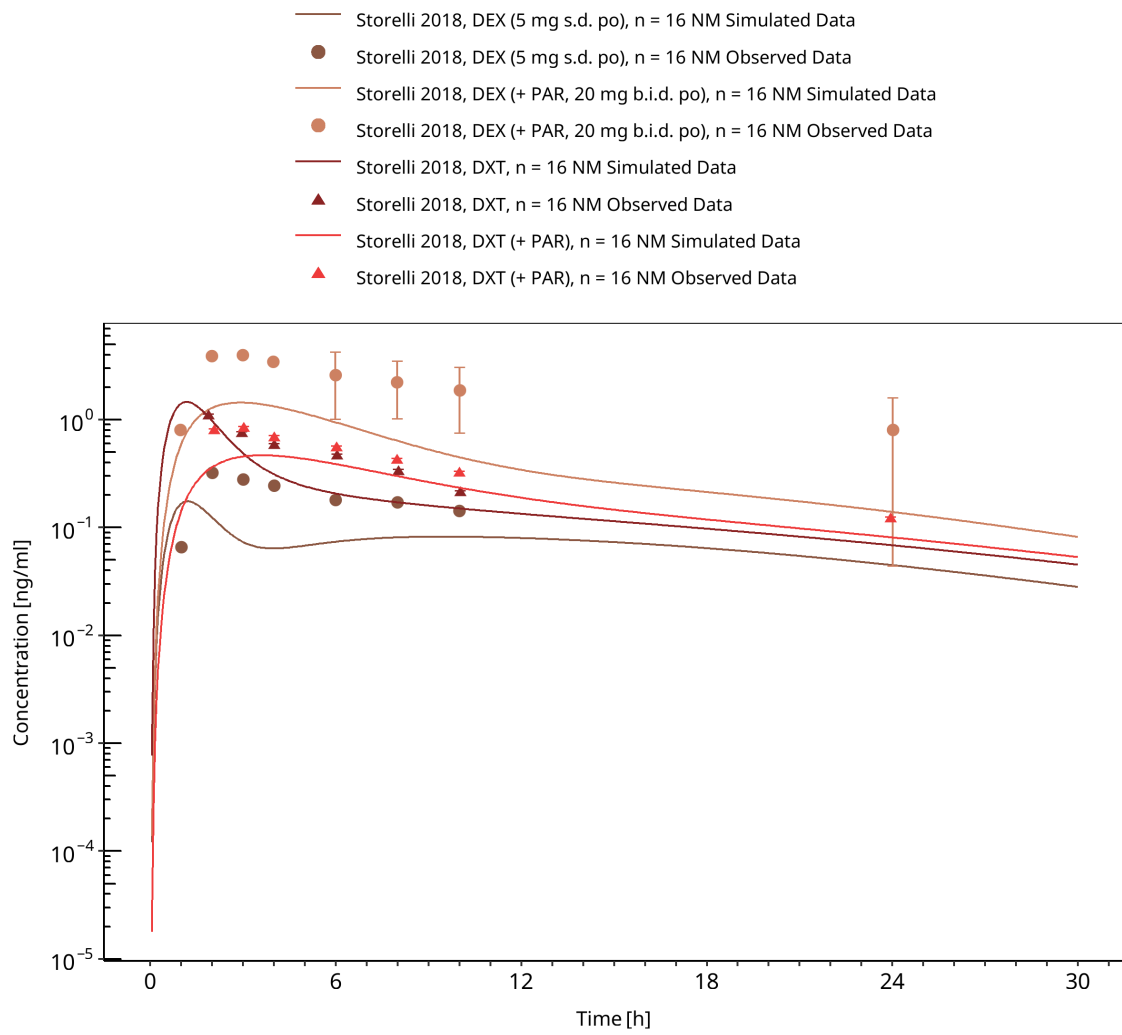


Figure 3-37: Paroxetine - Dextromethorphan DDGI

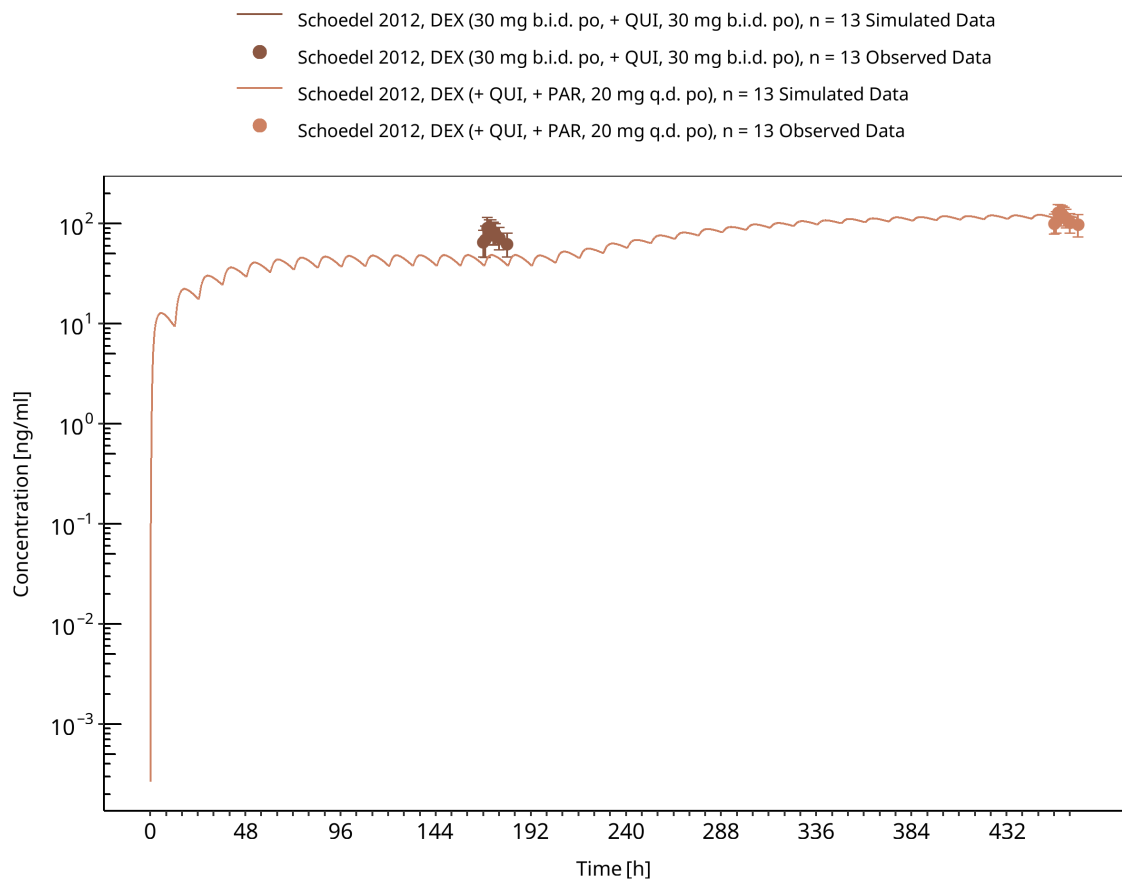


Figure 3-38: Paroxetine - Dextromethorphan DDI

4 Conclusion

The CYP2D6 DDGI network was qualified against clinical DD(G)I data for AUC and Cmax ratios across the selected perpetrator and victim combinations. In the current qualification run, the aggregate GMFE was 1.40 for AUC ratios and 1.38 for Cmax ratios. Most predictions were within two-fold of the observed values, with 86.32% of AUC ratios and 87.07% of Cmax ratios meeting this criterion.

The stricter limits proposed by Guest et al. were met by 57.26% of AUC ratios and 55.17% of Cmax ratios. The network therefore supports the intended use of the OSP Suite for CYP2D6-mediated DDI and DDGI simulations within the evaluated scenarios, while the remaining deviations indicate that model performance should be interpreted together with the underlying interaction mechanism, genotype group, victim drug, perpetrator drug, and analyte.

This qualification does not replace compound-specific model evaluation. New use cases should remain within the mechanistic and clinical-data domain covered by the qualified network or should be supported by additional verification.

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6 Appendix

6.1 Open Systems Pharmacology Suite (OSPS) Introduction

The Open Systems Pharmacology Suite provides a software environment for whole-body physiologically based pharmacokinetic modeling and simulation. The suite includes PK-Sim for model building and population simulation, MoBi for detailed model extension, and supporting R packages for qualification and reporting workflows.

Models are composed of physiological system information, compound-specific parameters, formulations, administration protocols, and observed data. Simulations can be combined with observed clinical data to evaluate whether model predictions are adequate for a defined context of use.

The qualification workflow uses versioned model snapshots and a machine-readable qualification plan to support reproducibility. The report generated from the plan documents the simulations, observed data comparisons, and summary performance metrics used to evaluate the intended use.

6.2 Mathematical Implementation of Drug-Drug Interactions

Drug-drug interaction simulations in the OSP Suite are represented through mechanistic changes in drug disposition processes. Depending on the compound and interaction scenario, these processes can include reversible enzyme inhibition, time-dependent inactivation, induction, transporter inhibition, transporter induction, or changes in multiple pathways.

For CYP2D6 DDGI scenarios, genotype-dependent activity is represented through phenotype-specific or activity-score-specific model settings where supported by the source model. The resulting simulations compare the victim-drug exposure with and without the perpetrator under the clinical dosing schedule. AUC and Cmax ratios are then calculated from matched control and interaction simulations.

The qualification plots compare predicted and observed exposure ratios. Performance is summarized using predicted versus observed plots, predicted/observed residual plots, geometric mean fold error, two-fold limits, and the limits proposed for ratio endpoints by Guest et al.

6.3 Automatic (re)-qualification workflow

The requalification workflow is driven by the qualification plan. The plan defines the model snapshots, simulations, observed datasets, output paths, comparison windows, figures, tables, and report sections.

The workflow loads the referenced model snapshots, runs the required simulations, calculates exposure ratios and concentration-time profile comparisons, and writes a report-ready output folder. The generated report should be reviewed for scientific consistency, expected model versions, complete figures, correct study metadata, and warnings from the simulation or reporting logs.

Before publication, the final report should be regenerated from a clean environment using the intended OSP Suite and qualification framework versions. The output should be checked against the committed qualification plan and the exact model snapshot releases used for qualification.

7 Glossary

| ALP | alprazolam | | AS | CYP2D6 activity score | | ATO | atomoxetine | | AUC | area under the plasma concentration-time curve | | AUClast | AUC calculated between first and last concentration measurement | | AUCR | AUC ratio | | b.i.d. | twice daily | | BUP | bupropion | | CBZ | carbamazepine | | CIM | cimetidine | | CLA | clarithromycin | | CLO | (E)-clomiphene | | Cmax | maximum plasma concentration | | CmaxR | Cmax ratio | | CYP | cytochrome P450 | | DDI | drug-drug interaction | | DDGI | Drug-drug-gene interaction | | DES | desipramine | | DEX | dextromethorphan | | DIG | digoxin | | DTT | total dextrorphan | | DXG | dextrorphan-O-glucuronide | | DXT | dextrorphan | | ERY | erythromycin | | FLV | fluvoxamine | | HDC | (E)-4-hydroxy-N-desethylclomiphene | | IM | CYP2D6 intermediate metabolizer | | ITR | itraconazole | | iv | intravenous | | KET | ketoconazole | | MET | metoprolol | | MEX | mexiletine | | MID | midazolam | | N | number of study participants | | NDC | (E)-N-desethylclomiphene | | NM | CYP2D6 normal metabolizer | | OHC | (E)-4-hydroxyclophene | | OHD | 2-hydroxydesipramine | | OHQ | 3-hydroxyquinidine | | OHR | 9-hydroxyrisperidone | | OME | omeprazole | | OSP | Open Systems Pharmacology | | PAR | paroxetine | | PBPK | physiologically based pharmacokinetic | | P-gp | P-glycoprotein | | PM | CYP2D6 poor metabolizer | | po | oral | | q.d. | once daily | | q.i.d. | four times daily | | QUI | quinidine | | RIF | rifampicin | | RIS | risperidone | | RME | (R)-metoprolol | | s.d. | single dose | | SME | (S)-metoprolol | | t.i.d. | three times daily | | VER | verapamil |